

Building a Quantum Black Hole Microstate: A Bulk Path Integral for a Heavy Virasoro Primary

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Abstract

Consider level- k $SL(2, \mathbb{R}) \times SL(2, \mathbb{R})$ Chern-Simons theory on a semi-infinite solid cylinder, with a Wilson line in the unitary principal continuous series inserted along the time direction at the center of the spatial disc. We argue that at the $\tau = 0$ cut, this Euclidean path integral prepares the bulk dual of a heavy Virasoro primary above the BTZ threshold, with Drinfel'd-Sokolov (DS) reduction playing a crucial role in converting the affine module into a Virasoro module. Semi-classically, the DS constraint turns into the DS gauge, familiar from asymptotically AdS_3 gravity. Together with boundary gravitons, these primaries provide a bulk construction of BTZ black hole microstates at finite k . In the large- k WKB limit, the saddle that controls an *individual* primary is the BTZ black hole with a *singular* horizon. We study the (grand-)canonical partition function of the microstates with density of states dictated by modular invariance. The temperature and chemical potential then fix the holonomy saddle to be the Euclidean BTZ black hole with a *smoothly* contractible thermal cycle. A density of heavy primaries growing as $e^{2\pi\gamma QP}$ would instead lead to a defect (or excess) saddle, except at the Cardy value $\gamma = 1$ fixed by modular invariance. After analytic continuation, the eternal Lorentzian black hole with a smooth horizon is therefore not just any thermal ensemble, but the “modular ensemble” of (semi-classically singular) horizonless microstates. We compute the thermal fluctuations in this ensemble, and reproduce the fluctuations in the area of the form expected from semi-classical general relativity. Ensembles of theories do not play a direct role in any of our discussions. We also note that these microstates, unlike fuzzballs, do *not* break the isometries of the black hole.

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1 Introduction

It is an oft-stated truism that a black hole is a thermal ensemble of heavy microstates. But the precise nature of these microstates in the bulk language is largely a mystery. The Gibbons-Hawking [1] calculation that is the gold-standard in producing black hole thermodynamics from the bulk, is *not* manifestly a count of the microstates.

It has been suggested [2]¹ that for generic finite temperature black holes, demanding a complete UV description of every microstate *in a given theory*, may not be necessary for a satisfactory understanding of black hole statistical mechanics. The “holographic atomic

¹See also motivating earlier work in [3, 4, 5].

hypothesis” advocated in [2] suggests a more hopeful possibility. Just as an atomic/molecular model can capture the microscopic origin of ordinary gas thermodynamics without being too sensitive to the specific UV completion, there may exist structures that are (semi-)universal across theories of quantum gravity and are useful for describing black hole microstates. In fact, the results of [2] indicate in a precise sense, that (at least in $\text{AdS}_3/\text{CFT}_2$) the situation is in fact *more* fundamental and universal for the thermodynamics of black holes, than for the thermodynamics of gases.

The key point is that in 2D CFTs with $c > 1$, one expects a bulk description of primaries via Wilson lines in a suitable Chern-Simons theory/TQFT. This construction is usually discussed semi-classically and below the BTZ threshold, often in connection with conical defects and geodesics. But as suggested in [2] and further developed in [6, 7] and in the present paper, it is a much more general structure. It implies that the nature of microstates is universal in $\text{AdS}_3/\text{CFT}_2$ across theories: what differs from theory-to-theory is the *specific* set of discrete primaries and their “interactions” as controlled by the OPE coefficients. In other words, the non-perturbative picture of the bulk is that of a TQFT skeleton [7] together with its sewing-compatible (discrete) bootstrap data.

While the ingredients are quite well-known, this perspective may not be – so let us explain this in a bit more detail. What we are suggesting is that we may already have a complete description of non-perturbative bulk AdS_3 quantum gravity, via *bulk* bootstrap in terms of a non-compact TQFT². A two-dimensional CFT naturally separates into universal kinematic data and theory-specific data. Virasoro representation theory fixes the characters, conformal blocks, and the kernels that relate different sewing channels. A particular CFT selects which primary representations actually occur and specifies their OPE coefficients. In the bulk, the first layer has a natural realization as a non-compact topological skeleton, closely related to Teichmüller or Virasoro TQFT [8, 9, 10, 11, 12, 13, 14, 2, 6, 7]. Wilson lines carry representation labels, Wilson networks compute conformal blocks, and changes of bulk decomposition implement the same sewing moves that underlie the CFT bootstrap on all surfaces with insertions. From this perspective, the Wilson line is not merely a useful probe inserted into an already existing geometry. It is the natural topological carrier of the quantum numbers of a primary state.

The Chern-Simons/TQFT formulation allows the Euclidean horizon to be replaced by an inner boundary carrying holonomy data, and the modular decomposition of the BTZ character can then be interpreted as a sum over bulk states which are not individually

²We emphasize that *explicitly* solving for sewing compatible bootstrap data is a tall order. So while we do believe this is a powerful re-frame, it is not the same things as the *ability* to solve explicit theories. This picture should be compared to that in rational CFTs, where the bulk (compact) TQFTs are completely solvable and the number of primaries (of some extended chiral algebra) is finite.

described by the smooth Euclidean black-hole geometry [2]. The smooth horizon partition function appears only after the appropriate spectral sum, as was missing from the Gibbons-Hawking calculation. In the Cardy regime [15], the discrete sum over explicit microstates in a specific holographic CFT is extremely well-approximated by the continuum sum over the S -kernel density of states [2]. This can be viewed as the “smooth horizon” approximation.

These ideas naturally suggest a concrete bulk microscopic interpretation of heavy CFT primaries. Once an individual compact CFT contains a primary with given $(h, \bar{h})$, the topological skeleton should contain a corresponding bulk state labeled by the associated Wilson line. The skeleton by itself does not determine which such primaries occur in the exact theory, nor their multiplicities. Those are part of the theory-specific CFT data. What it does provide is a universal bulk realization of the state associated with each allowed primary. Below the BTZ threshold this statement is familiar from conical defects, which are associated with elliptic holonomy and discrete-series representations. Above threshold the analogous object has hyperbolic holonomy. This points to Wilson lines in the principal continuous series as the natural bulk representatives of heavy Virasoro primaries.

One purpose of this paper is to make this statement explicit at the level of the bulk state. We consider $SL(2, \mathbb{R}) \times SL(2, \mathbb{R})$ Chern-Simons theory on a semi-infinite solid cylinder and insert a Wilson line along the Euclidean time direction at the center of the spatial disc. The Euclidean path integral then prepares a state on the $\tau = 0$ cut. The state first lives in an affine current-algebra module. The physical gravitational state is obtained only after imposing the reduction associated with the asymptotically AdS_3 boundary conditions. In the language used below, this is a quantum Drinfel’d-Sokolov reduction. The result is a Virasoro primary together with its boundary-graviton descendants. Thus the Wilson-line state provides a bridge between the finite- c TQFT skeleton and the more familiar bulk description in which a semiclassical bulk geometry can be extracted from the gauge field.

A fixed heavy primary is a state with a definite Wilson-line label and definite hyperbolic holonomy (but with no a priori thermal periodicity). In the large- k limit its saddle is the BTZ gauge field, but with the Wilson-line source. In the metric language, a semi-classical primary is therefore naturally interpreted as having a singular horizon, a place where “space-time ends” in the sense of geodesic incompleteness³. Smoothness emerges after summing over heavy primary sectors. In the Cardy regime, modular invariance fixes the universal continuum approximated density through the vacuum row of the modular S -kernel. Extremizing this modularly weighted partition sum over the Wilson-line label selects precisely the holonomy for which the Euclidean thermal cycle becomes smoothly contractible. In other words, the Gibbons-Hawking regularity condition is an emergent saddle-point statement about the

³Note that geodesics are probes that only make sense on coherent heavy backgrounds, so on a microstate, we are using it only as an intuitive tool rather than as a physical probe.

spectrum rather than a microscopic condition imposed on every state.

The resulting picture is the one we will develop throughout the paper. The fundamental finite- c objects are quantum states, while a smooth black-hole geometry is a collective semiclassical description of an appropriate set of such states. The exact spectrum of an individual CFT remains discrete. The modular kernel supplies its universal continuum description in the regime in which Cardy coarse graining is valid. No ensemble of theories is required. What emerges is instead a “modular ensemble” of states within a theory. In this language, the holographic atomic hypothesis comes into its own: the relevant atoms are heavy Wilson-line sectors, and the smooth BTZ horizon is a collective property that emerges from them. Our first task is therefore to explain how the Wilson-line state is prepared and how its affine current-algebra description is converted into the physical Virasoro microstate.

1.1 Strategy: Quantize-then-Reduce

The central technical ingredient in this paper is a two-step construction. A Wilson line along the Euclidean time axis first prepares a state in an *unreduced* affine $\widehat{\mathfrak{sl}}(2, \mathbb{R})_k$ module on the punctured spatial disc. The one-unit spectrally flowed Drinfel’d-Sokolov (DS) reduction appropriate to the Brown-Henneaux boundary conditions on the cylinder [16, 17] is then used to pass from this affine module to the physical Virasoro module. Since the unreduced quantum $SL(2, \mathbb{R})$ Chern-Simons theory is subtle⁴ – the gauge group is non-compact and a satisfactory quantization is not available to our knowledge – we first orient the reader on how this strategy relates to previous approaches, in which the reduction appears in different guises.

In the Wilson-line computations of Virasoro blocks [18, 19, 20, 21, 22], the reduction enters as a renormalization condition rather than as an operation on a Hilbert space. Matrix elements of the $\mathfrak{sl}(2)$ Wilson line are divergent at the quantum level, and the ambiguities are fixed by demanding consistency with *Virasoro* (rather than current-algebra) Ward identities. The resulting anomalous dimension shifts the naive weight to its DS/Liouville value. The choice of which Ward identities define the quantum theory can be viewed as the place where reduction enters⁵, imposed order by order in $1/c$, even though the constraint itself is never exhibited acting on a module. In the coadjoint-orbit approach [23, 24, 25, 26], the reduction is instead performed classically, before quantization: one starts from the already-reduced phase space – Virasoro coadjoint orbits, equivalently Bañados geometries with Brown-Henneaux

⁴Note that this correlates with the fact that pure 3D Einstein gravity is likely not the best starting point for a unitary quantum gravity.

⁵While we believe this is natural, it should be emphasized that this is best viewed as an *interpretation* of the calculation.

boundary conditions – and quantizes that. This route never encounters the unreduced quantum theory at all. Finally, in the Virasoro TQFT of [13, 14], the reduction is built into the very definition of the theory: the Hilbert spaces are *defined* to be spaces of Virasoro conformal blocks, so that the outcome of the reduction is taken as an axiom, and defects enter directly as Virasoro primary labels. In none of these approaches is the unreduced affine puncture sector on the disc written down. Our attitude is that the affine module is nevertheless useful as an intermediate “upstairs” description: we never rely on a full quantization of $SL(2, \mathbb{R})$ Chern-Simons theory, only on the fixed Wilson-line sector, which we do eventually reduce.

The utility of this “upstairs” object is that it allows us to make connections with semi-classical *bulk* solutions in the large- k limit. We find that this gives clarity on multiple aspects of the physics: specifically in matters of dynamics, semi-classics and bulk geometry. First, the construction is Hartle-Hawking-like. An explicit Euclidean bulk path integral with a Wilson-line source prepares the unreduced affine state on the $\tau = 0$ cut (and the flowed Brown-Henneaux reduction then gives the physical Virasoro state). The reduce-then-quantize route hands one the Hilbert space, but has no counterpart of this bulk preparation. Second, the semi-classical discussion of Sections 4 and 5 – the WKB limit of the wavefunctional, the saddle-point connection, and the resulting singular-versus-smooth horizon geometries – is a statement about bulk saddles of the Chern-Simons theory and allow a natural bulk metric connection. The coadjoint-orbit description is intrinsically a boundary theory, with no bulk interior whose smoothness can be interrogated. Keeping the connection as the basic variable is what allows us to ask questions about the horizon. Third, the bulk mechanism that distinguishes the sub- and super-threshold sectors – elliptic holonomy and conical defects versus hyperbolic holonomy and black hole microstates – is more intuitive upstairs, whereas downstairs they are all encoded as normal orbits.

Our approach also provides us a comparison⁶ between quantize-then-reduce and reduce-then-quantize. Let us explain this. For the sub-threshold (discrete-series) sector of Section 2, the affine module is of lowest-weight type⁷. For the unflowed plane reduction this lies within the standard highest-weight quantum DS framework [27, 28, 29, 30, 31, 32]. The gravitational problem considered here, however, is the one-unit spectrally flowed cylinder reduction, so we do not invoke the standard unflowed reduction theorems as an all-level proof. The super-threshold (continuous-series) sector of Section 3 has the additional feature that the affine module is relaxed⁸, with no highest- or lowest-weight vector, and hence lies still further

⁶Or perhaps a consistency check.

⁷A lowest-weight module is a highest-weight module for the opposite Borel subalgebra, so we use the two terms somewhat interchangeably.

⁸The terminology on affine representations is a bit baroque in the mathematics literature. What we call a relaxed module induced from the principal continuous series seems to be called (extremely confusingly!)

outside the usual extremal-weight setting. For the principal series the level-zero mechanism can nevertheless be analyzed directly: at fixed s and fixed ϵ , the cylinder constraint selects a single Virasoro primary class, as spelled out in Appendix B. The positive-level BRST cohomology of the flowed discrete and relaxed modules will be discussed in [33]⁹. The all-level structure – for generic super-threshold momentum, a single non-degenerate Verma module built on a primary with $h > \frac{c-1}{24}$ – is also what reduce-then-quantize produces from the quantization of the hyperbolic $\text{Diff}(S^1)/S^1$ coadjoint orbit [23, 24, 26, 2]. We therefore view the two routes as complementary: the orbit quantization corroborates the reduced Hilbert space, while the Wilson-line construction supplies what the orbit description cannot – the bulk preparation of the microstate and its saddle-point geometry.

1.2 Microstates and the Nature of the Smooth Horizon

In the second half of the paper (sections 4 and 5) we discuss the relationship between individual black-hole microstates and a smooth black-hole geometry. A heavy primary defines a perfectly good quantum state, represented in the bulk by a Wilson line whose conjugacy class fixes the BTZ parameters. In the large- k limit, the corresponding geometry is locally BTZ and contains a finite-perimeter would-be horizon. However, the state-preparation path integral does not impose the thermal identification required for smoothness of the Euclidean horizon. The semiclassical (i.e., large- k) geometry of an individual microstate therefore ends at a singular finite-perimeter surface, rather than extending through a smooth Kruskal horizon. Note that this singularity of the microstate is a semi-classical artifact – at finite- k it is a well-defined quantum state.

The smooth geometry emerges only after summing over heavy states. In the Cardy regime, modular invariance fixes the asymptotic density of primaries through the vacuum row of the Virasoro S -kernel. Since the integration variable is also the Wilson-line label, the thermal sum is directly a sum over bulk holonomies. At large k , the competition between the Cardy density and the Boltzmann weight gives

$$\lambda_L = \frac{2\pi}{\beta_L}, \quad \lambda_R = \frac{2\pi}{\beta_R}, \quad (1.1)$$

which are precisely the conditions for smoothness of the Euclidean BTZ horizon. The Gibbons–Hawking regularity condition is therefore not a property imposed on each microstate, it appears as the saddle-point condition of the thermal sum.

a “relaxed highest weight” representation in some places. We will not follow that nomenclature. As a mnemonic: we will have relaxed above threshold and highest/lowest-weight below.

⁹Spectrally flowed DS reductions are discussed from a mathematical perspective in [34].

This also shows why *any* thermal ensemble in itself is not sufficient. If the Cardy growth exponent of the primary density is rescaled by a factor γ , the saddle instead gives

$$\lambda_{L,R}\beta_{L,R} = 2\pi\gamma. \quad (1.2)$$

The resulting geometry has a conical defect or excess at the horizon, and simultaneously $S = \gamma \frac{A}{4G_N}$. Thus smoothness and the Bekenstein–Hawking relation are tied to the same microscopic input: the modularly determined growth of states. Modular invariance supplies the spectral weighting required for the thermal state to have a smooth semiclassical horizon. In Lorentzian signature, although a thermofield-double purification can be written for any thermal density matrix, a smooth Einstein–Rosen bridge and Kruskal extension naturally require such a *modular ensemble*.

The entropy itself has a similarly direct interpretation. The S -kernel gives the Cardy growth of heavy Virasoro primaries, while boundary-graviton descendants provide the remaining contribution, and together they reproduce the full Bekenstein–Hawking entropy [2]. The horizon area is therefore associated with the dominant holonomy in a statistical ensemble of Wilson-line states.

Finally, thermal fluctuations around the saddle are fluctuations of the same Wilson-line label, or equivalently of the horizon holonomy. They give

$$\text{Var}(S) = S, \quad \text{Var}(A) = 4G_N A, \quad \frac{\Delta A}{A} \sim S^{-1/2}. \quad (1.3)$$

The smooth horizon is consequently the sharply peaked, large-entropy limit of this distribution.

The resulting picture is that heavy primaries furnish the microscopic BTZ states, but smoothness of the horizon is a collective property of the thermal ensemble with a modular density of states.

1.3 Comments

Let us conclude the Introduction with some technical and conceptual observations.

- The Drinfel’d-Sokolov reduction we do in this paper naturally leads us to a central charge for the vacuum AdS_3 that is consistent with the “regulate-and-quotient” prescription of [2]. This strengthens the case¹⁰ that the classical central charge C of the orbit and the quantum central charge are related by $c = C + 1$ across orbits. This is in contrast to the special treatment for the exceptional orbit that is sometimes suggested, with $c = C + 13$. We discuss this in more detail in Section 2.5.2.

¹⁰See [35] for an earlier calculation that naturally enables this interpretation.

- In the TQFT skeleton language, the construction of the Virasoro primary (and the associated Verma module) from the Wilson line is essentially tautological. This is because the Wilson labels there are automatically those of Virasoro and not affine. But the connection with semi-classical gravity and the metric description is clearer in the Chern-Simons language. Apart from allowing us to connect with the familiar BTZ language and the role of smooth and unsmooth horizons, the Chern-Simons/DS reduction also clarifies the role of Brown-Henneaux boundary conditions from a quantum-first perspective.
- A feature of the construction that was noted in [2, 36] is that hyperbolic holonomy Wilson line sources lead to metric configurations where the source has a “size”, equal to the horizon size of the associated black hole. We do not believe there is any contradiction here: the Wilson line is placed at the origin of the coordinate disc, and is a topological object, while size is a metric notion. (Hyperbolic) holonomy in one language translates to horizon size in the other. This simple yet remarkable observation does not seem to be as widely appreciated as it should be.
- Another fact, which was again mentioned in [2] is that these microstates do not break the isometries of the BTZ black hole. This is satisfying because in many BPS index computations, it is known that the microstates that contribute to a black hole’s index preserve its isometries [37, 38, 39]. This is sometimes used as an argument against fuzzball constructions [40, 41] that break the symmetries of the horizon [38, 42].
- The modular ensemble perspective allows us to understand why the smooth horizon is such a persistent and compelling feature of black hole physics, despite the microstates (in the semi-classical limit) being singular at the horizon. The key point is that the Cardy regime is a well-defined regime even at finite- c . So vacuum dominance in the dual channel and utility of the S -kernel remain intact as valid approximations even away from the semi-classical limit. Some simple calculations that quantify this point, will be presented in a companion paper [43].
- We mentioned earlier in a footnote that geodesics are not the best tools to characterize semi-classical microstates, because they are naturally defined on coherent heavy backgrounds. More fundamentally, we believe that while we can write down a metric that is translated from the Chern-Simons field in this (large- k) limit, the *physical* significance of the metric language itself is questionable. The semi-classical microstate is perhaps best left as a Chern-Simons saddle with a source. In our discussions in Section 4, such caveats about the *utility* of individual semi-classical microstates should be kept in mind. While it is instructive to study such objects for some purposes, the

dynamics of a light probe interacting with the heavy microstate is a detailed question that also involves OPE coefficients (and ETH, and such). Our eigen-microstates enable such questions to be meaningfully asked, but we leave them for the future. Given our findings, it is a natural possibility that a smooth interior will emerge from the dynamics/interaction of the probe (say, a light primary) with a single heavy microstate [44, 45], when the probe has only coarse-grained resolution: this is when we expect that the heavy microstate can be replaced by the ensemble of horizonless microstates that act like a smooth horizon. One suspects that this, is what underlies infall.

2 Conical Defect Microstates and Sub-Threshold Primaries

Before turning to the black-hole regime, it is useful to recall the sub-threshold version of the construction. A conical defect in AdS_3 is characterized by an elliptic holonomy around the particle worldline. In Chern-Simons language this holonomy is naturally implemented by a Wilson line piercing the spatial disc. The Wilson line label specifies a sector of the bulk Hilbert space, and the Euclidean path integral on a half-cylinder prepares the corresponding representation-valued affine state. Keeping the open Wilson-line index free retains the whole horizontal¹¹ multiplet, see Appendix A. Contracting it with an endpoint vector chooses an upstairs (affine) representative. This choice will not survive as a multiplicity of gravitational primaries after the Drinfel’d-Sokolov reduction.

The conical-defect case is the simpler sub-threshold analogue of the construction developed in Section 3. It illustrates three structural features that will persist above the BTZ threshold:

1. a Wilson line on the time axis turns the spatial disc into a punctured disc;
2. the puncture label fixes the conjugacy class of the linking spatial holonomy;
3. after imposing the gravitational boundary conditions – the Drinfel’d-Sokolov reduction described in Section 2.5 – the horizontal affine data reduce to a Virasoro primary and, at positive level, the expected Virasoro descendant tower. We will present the general structure here, but the detailed all-level BRST cohomology is left for [33].

We use the phrase “fixed conical defect” to mean the primary label, or holonomy conjugacy class. Descendants are boundary-graviton dressings of this primary sector. They carry

¹¹Here and below, “horizontal” refers to the ordinary $\mathfrak{sl}(2, \mathbb{R})$ subalgebra generated by the affine zero modes J_0^a . A horizontal representation is therefore the representation carried by the level-zero states before the nonzero current modes act.

energy: Virasoro descendants have $L_0 = h + N$ and $\bar{L}_0 = \bar{h} + \bar{N}$, but they share the same primary/holonomy label.

2.1 AdS₃ Gravity from Chern-Simons Theory

We begin by fixing conventions. Three-dimensional Einstein gravity with negative cosmological constant $\Lambda = -1/\ell^2$ can be written as a difference of two Chern-Simons theories [46, 47]. At the Lie-algebra level the gauge symmetry is $\mathfrak{sl}(2, \mathbb{R})_L \oplus \mathfrak{sl}(2, \mathbb{R})_R$ ¹². The two gauge fields are

$$A = \left(\omega^a + \frac{e^a}{\ell} \right) J_a, \quad \tilde{A} = \left(\omega^a - \frac{e^a}{\ell} \right) J_a. \quad (2.1)$$

Here e^a is the dreibein, ω^a is the spin connection, and J_a are generators of $\mathfrak{sl}(2, \mathbb{R})$ satisfying

$$[J_a, J_b] = \epsilon_{abc} \eta^{cd} J_d, \quad \eta^{ab} = \text{diag}(-1, +1, +1). \quad (2.2)$$

J_a denotes a basis of the finite-dimensional classical Lie algebra. In the Chern-Simons/gravity formulas, Tr denotes the fundamental-representation trace. For the Cartan–Weyl basis used below we choose

$$H = \frac{1}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad E = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad F = \begin{pmatrix} 0 & 0 \\ -1 & 0 \end{pmatrix},$$

so that

$$[H, E] = E, \quad [H, F] = -F, \quad [E, F] = -2H,$$

and

$$2\text{Tr}(HH) = 1, \quad 2\text{Tr}(EF) = -2.$$

The affine current algebra below uses the same Lie-algebra basis and commutation relations, $(J^3, J^+, J^-) \equiv (H, E, F)$. But there, the invariant form will be the opposite: $\kappa(X, Y) \equiv -2\text{Tr}(XY)$, with $\kappa^{33} = -1$ and $\kappa^{+-} = 2$. We reserve J_n^a for the corresponding affine-current modes. The action, which is equal to the Einstein action up to boundary terms, is

$$S = S_{\text{CS}}[A] - S_{\text{CS}}[\tilde{A}], \quad S_{\text{CS}}[A] = \frac{k}{4\pi} \int_{\mathcal{M}} \text{Tr} \left(A \wedge dA + \frac{2}{3} A \wedge A \wedge A \right), \quad (2.3)$$

with

$$k = \frac{\ell}{4G_N}. \quad (2.4)$$

At leading semiclassical order the Brown-Henneaux central charge is $c = 6k$. We will use this leading relation for semiclassical matching.

¹²Throughout, our Euclidean continuation is the coordinate Wick rotation $t = -i\tau$ of the Lorentzian Chern-Simons description, with the noncompact gauge algebra and Wilson-line representation labels kept fixed. This follows Section 2 of [48] and Appendix B of [2]. At the level of the group, we use the universal cover, so the discrete-series label j need not be quantized.

2.2 Conical Defects and Elliptic Holonomy

A spinless massive particle at the centre of AdS_3 produces a conical defect when its mass lies below the BTZ threshold. A convenient Euclidean form of the metric is

$$ds^2 = (r^2 + \alpha^2) d\tau^2 + \frac{dr^2}{r^2 + \alpha^2} + r^2 d\phi^2, \quad \phi \sim \phi + 2\pi, \quad (2.5)$$

where $0 < \alpha \leq 1$. The deficit angle is $2\pi(1 - \alpha)$. The endpoint $\alpha = 1$ is global AdS_3 , while $\alpha \rightarrow 0^+$ approaches the BTZ threshold from below. One definition of the ADM mass is

$$8G_N M = -\alpha^2,$$

and the corresponding left and right conformal weights of the spinless primary are

$$h_L = h_R = \frac{c}{24}(1 - \alpha^2). \quad (2.6)$$

Thus $h = 0$ for global AdS_3 and $h \rightarrow c/24$ at the BTZ threshold.

The same distinction is expressed gauge-theoretically by the holonomy around a small spatial circle linking the particle worldline¹³. For a conical defect this holonomy lies in an elliptic conjugacy class. In the fundamental representation one may write

$$\text{Hol}(A) \sim \begin{pmatrix} e^{i\pi\alpha} & 0 \\ 0 & e^{-i\pi\alpha} \end{pmatrix}, \quad \text{Tr}_2 \text{Hol}(A) = 2 \cos(\pi\alpha). \quad (2.7)$$

The matrix in (2.7) should be understood as a complex-diagonal representative of the real elliptic conjugacy class of $SL(2, \mathbb{R})$. The corresponding statement for the right-moving connection is analogous (including for spinning conical defects).

2.3 Wilson Lines as Sources

A massive particle in Chern-Simons theory is naturally described by a Wilson line supported on its worldline. Let γ be the line $\{0\} \times \mathbb{R}_\tau$ at the centre of the spatial disc. A Wilson line in a representation R of $SL(2, \mathbb{R})$ may be denoted schematically as

$$W_R(\gamma) = \text{Tr}_R \mathcal{P} \exp \left(\int_\gamma A \right), \quad (2.8)$$

or, more precisely for an open line ending on a time slice, by the corresponding worldline path integral with endpoints in the representation space of R .

¹³We will often use the phrase “particle worldline” and “Wilson line” interchangeably. In the spinless elliptic (conical defect) case, in a suitable semi-classical limit, the two are the same. But the geodesic limit will not be our concern in this paper.

The physical datum carried by the line is the conjugacy class of the holonomy of the bulk connection around a small circle linking the line. Equivalently, at fixed Euclidean time the spatial slice is $D^2 \setminus \{0\}$, and the Wilson line labels the puncture at the origin. In the sub-threshold case this linking holonomy is elliptic, in the super-threshold case it will be hyperbolic.

The coadjoint-orbit description is a convenient way to describe this [49, 50, 51]. Even though we will not need it much, the intuition is useful and so we provide a quick summary of some facts. The idea here is that one can write the Wilson line above (2.8) using an auxiliary path integral over a group-valued field on the worldline. Concretely, one introduces a field $U(s) \in SL(2, \mathbb{R})$ on the worldline with first-order action $\int ds \operatorname{Tr}(\mu U^{-1} D_s U)$, where μ is a fixed Lie-algebra element encoding the mass and spin of the particle, and D_s is the covariant derivative along the worldline. Because the action is first order, it defines a phase space rather than a configuration space: the physical phase space variable is a point on the “co-adjoint orbit” of μ :

$$Q^{\text{orb}} = U\mu U^{-1} = q^a J_a \in \mathfrak{sl}(2, \mathbb{R}),$$

where the coadjoint charge has been identified with a Lie-algebra element using the invariant bilinear form appearing in the Chern-Simons action.¹⁴ The charge lives on the set of all conjugates of μ – which is the coadjoint orbit $\mathcal{O}_\mu$ of $SL(2, \mathbb{R})$. The internal degree of freedom of the particle is thus a classical mechanical system on $\mathcal{O}_\mu$, coupled to the pullback of the Chern-Simons connection through the term linear in A_s . Quantizing this orbit – which can be implemented by the path integral over U – gives a representation R . The conjugacy class of Q^{orb} is fixed by μ , and upon quantization its components q^a are promoted to the generators of R . The equation of motion in the presence of the line source is

$$F_{ij}(x) = \frac{2\pi}{k} q^a \delta^{(2)}(x) \epsilon_{ij} J_a, \quad (2.9)$$

so the connection is flat away from the puncture. On the punctured disc one may write, up to conjugacy,

$$A_\phi = \frac{1}{k} Q^{\text{orb}} \sim \frac{1}{k} \mu, \quad \operatorname{Hol}(A) = \exp(2\pi A_\phi) = \exp\left(\frac{2\pi\mu}{k}\right). \quad (2.10)$$

Thus the orbit label μ – equivalently, the representation data of the Wilson line – determines the conjugacy class of the linking holonomy.

¹⁴This trace pairing fixes the normalization of the classical source and holonomy. The operator Casimir below is written in a Cartan–Weyl convention adapted to the quantum representation. We therefore keep the classical orbit radius and the quantum label j distinct except in their semiclassical matching.

2.4 The Unreduced Conical-Defect Sector

We now discuss the Wilson-line construction. The Wilson line first prepares a puncture sector of the unreduced $\widehat{\mathfrak{sl}}(2, \mathbb{R})_k$ current algebra. Brown-Henneaux boundary conditions are imposed by Drinfel'd-Sokolov (DS) reduction. The Virasoro primary and its boundary-graviton tower are therefore properties of the *reduced* state space, not labels that have to be supplied to the Wilson line from the outset.

2.4.1 Discrete Series and its Affine Module

For an elliptic Wilson line the relevant horizontal representations are the lowest-weight discrete series of $SL(2, \mathbb{R})$. We use the Cartan–Weyl basis $\{J^3, J^\pm\}$,

$$[J^3, J^\pm] = \pm J^\pm, \quad [J^+, J^-] = -2J^3. \quad (2.11)$$

The representation $\mathcal{D}_j^+$ is generated from an extremal vector $|j, j\rangle$ obeying

$$J_0^3 |j, j\rangle = j |j, j\rangle, \quad J_0^- |j, j\rangle = 0, \quad (2.12)$$

and has J_0^3 spectrum $j, j+1, j+2, \dots$. We work with the universal cover, so j need not be quantized. With the Casimir convention

$$\widehat{C}_2 = -(J_0^3)^2 + \frac{1}{2}(J_0^+ J_0^- + J_0^- J_0^+), \quad C_2(\mathcal{D}_j^+) = -j(j-1), \quad (2.13)$$

the Sugawara weight of every affine-primary state in this horizontal multiplet is

$$\Delta_j = \frac{C_2(\mathcal{D}_j^+)}{k-2} = -\frac{j(j-1)}{k-2}. \quad (2.14)$$

The unreduced affine module is obtained by acting with the negative current modes,

$$\widehat{\mathcal{D}}_j^+ = \text{span}\{(J_0^+)^p J_{-n_1}^{a_1} \cdots J_{-n_q}^{a_q} |j, j\rangle ; p \geq 0, n_i > 0\}. \quad (2.15)$$

The infinite horizontal tower is part of the unreduced Chern-Simons state space.

2.4.2 The Half-Cylinder State

At fixed Euclidean time the spatial slice is the punctured disc $D^2 \setminus \{0\}$, with the puncture labelled by $(j, \bar{j})$. Before reduction the corresponding chiral state spaces are

$$\mathcal{H}_{\text{aff}}(D^2; j, \bar{j}) = \widehat{\mathcal{D}}_j^+ \otimes \widehat{\mathcal{D}}_{\bar{j}}^+. \quad (2.16)$$

For a spinless defect we take $j = \bar{j}$.

The Euclidean path integral on $M_- = D^2 \times (-\infty, 0]$, with a Wilson line running along the Euclidean-time direction and ending on the puncture of the final slice, prepares the corresponding Chern-Simons state. At the asymptotic cylinder we use the standard chiral Chern-Simons boundary term and polarization for which the unreduced boundary degrees of freedom carry the affine current algebra. Brown-Henneaux boundary conditions are *not* yet imposed. Schematically,

$$|\Psi_{j,\bar{j}}\rangle_{\text{aff}} = \int_{\mathcal{P}_{\text{aff}}} [\mathcal{D}A] [\mathcal{D}\tilde{A}] W_{\mathcal{D}_j^+}[A] W_{\bar{\mathcal{D}}_{\bar{j}}^+}[\tilde{A}] e^{-S_E[A,\tilde{A}]} |A, \tilde{A}\rangle_{\tau=0}, \quad (2.17)$$

where S_E denotes the Euclidean continuation of (2.3), including the boundary term. The two Wilson factors refer to the independent left and right Chern-Simons sectors. The bar on $\bar{\mathcal{D}}_{\bar{j}}^+$ labels the right-moving copy, and no complex conjugation of the gauge field is implied.

Throughout the barred copy we use the opposite Cartan–Weyl basis $(\bar{J}^3, \bar{J}^+, \bar{J}^-) \equiv (-H, F, E)$. Thus the superscript $+$ on $\bar{\mathcal{D}}_{\bar{j}}^+$ denotes lowest weight with respect to this barred basis, in particular $\bar{J}_0^- |\bar{j}, \bar{j}\rangle = 0$. The barred basis obeys the same abstract commutation relations as (2.11), with the same invariant form -2Tr , it also has $\bar{\kappa}^{33} = -1$ and $\bar{\kappa}^{+-} = 2$, so the affine conventions are identical in form.

Keeping the endpoint representation index open, the puncture supplies the whole horizontal multiplet of affine-primary states. For each $v \in \mathcal{D}_j^+$ there is a level-zero state $|j; v\rangle \in \widehat{\mathcal{D}}_j^+$ satisfying

$$J_{n>0}^a |j; v\rangle = 0, \quad (2.18)$$

and similarly in the barred sector. Contracting the open endpoint with a particular vector v chooses one such upstairs state. Leaving the index open keeps the construction representation-valued. The Brown-Henneaux reduction below removes this horizontal degeneracy and selects the gravitational primary cohomologically.

2.4.3 Semi-Classical Matching with the Deficit Angle

The discrete-series label also fixes the classical elliptic holonomy. To match the complex-diagonal form of the elliptic holonomy in (2.7), and to make its relation to the hyperbolic sector above threshold transparent, it is convenient to place the corresponding factor of i in the elliptic generator. We therefore define

$$H_{\text{ell}} \equiv iH = \frac{i}{2}\sigma_3$$

and parametrize the classical orbit charge as $Q_{\text{ell}}^{\text{orb}} = 2\rho H_{\text{ell}}$. The linking holonomy is then

$$\text{Hol}(A) = \exp\left(\frac{2\pi}{k} Q_{\text{ell}}^{\text{orb}}\right) = \begin{pmatrix} e^{2\pi i \rho/k} & 0 \\ 0 & e^{-2\pi i \rho/k} \end{pmatrix}. \quad (2.19)$$

Matching this with (2.7) gives the classical relation $2\rho/k = \alpha$. Upon quantization, the coadjoint orbit becomes the representation carried by the Wilson line, so its classical orbit parameter ρ is correspondingly mapped to the representation label j . Standard coadjoint-orbit quantization makes this relation precise: $\rho = j - \frac{1}{2}$ [52]. Since the match to the geometry is by definition semiclassical, the relation we will use is

$$j = \frac{k\alpha}{2} + O(1). \quad (2.20)$$

In particular, $j = k/2$ approaches the global AdS_3 endpoint $\alpha = 1$ in the large- k limit. The corresponding Brown-Henneaux weight inferred from the classical geometry is

$$h_{\text{grav}} = \frac{c}{24}(1 - \alpha^2) = \frac{k}{4} - \frac{j^2}{k} + O(1), \quad (2.21)$$

where we treat $j \sim k$. We now derive the finite- k counterpart of this discussion directly from the quantum constraint.

2.5 Brown-Henneaux Reduction to Virasoro

The reduction relevant for gravity is the one in the boundary *cylinder* frame. This is important because the DS/Brown-Henneaux constraint on the cylinder and the frequently used constant DS constraint on the plane correspond to different affine modes. They are related by one unit of spectral flow.

2.5.1 The Cylinder Constraint and Spectral Flow

Brown-Henneaux boundary conditions are imposed directly on the boundary cylinder. In the usual lowest-weight DS gauge they fix the coefficient of F in

$$a_\phi = F - \frac{2\pi}{k} \mathcal{L}(\phi) E \quad (2.22)$$

to a nonzero constant, while the remaining field $\mathcal{L}(\phi)$ is the classical Virasoro stress tensor [16, 26]. In the quantum current algebra, the classical F direction becomes the affine J^- current¹⁵. Since J_n^a are the Fourier modes of the boundary current, this immediately gives

$$J_{\text{cyl}}^-(w) = 1 \quad \Longleftrightarrow \quad J_n^- = \delta_{n,0}. \quad (2.23)$$

We have chosen the normalization to be unity. Its magnitude can be absorbed by a constant Cartan rescaling. To compare with the standard plane discussion, let $w = \tau + i\phi$ and $z = e^w$. Before the DS improvement the affine current has weight one, so

$$J_{\text{pl}}^-(z) = \sum_{n \in \mathbb{Z}} J_n^- z^{-n-1}, \quad J_{\text{cyl}}^-(w) = z J_{\text{pl}}^-(z) = \sum_{n \in \mathbb{Z}} J_n^- e^{-nw}. \quad (2.24)$$

¹⁵Note that the invariant form we use here has the opposite sign, compared to the one in Section 2.1.

Thus the commonly used constant plane condition $J_{\text{pl}}^-(z) = 1$ fixes $J_{-1}^- = 1$, while the Brown-Henneaux cylinder condition fixes $J_0^- = 1$.

In modes the first-class constraints are therefore

$$\varphi_n = J_n^- - \delta_{n,0} \approx 0, \quad n \in \mathbb{Z}. \quad (2.25)$$

For the right-moving copy the identical-looking condition is $\bar{J}_n^- = \delta_{n,0}$ in the barred basis above. In the common matrix basis this fixes the E component of $\tilde{a}_\phi$, since $\bar{J}^- = E$, while the remaining component carrying $\tilde{\mathcal{L}}$ is $\bar{J}^+ = F$. Since $[J_m^-, J_n^-] = 0$, their algebra is abelian without a quantum central term. The associated BRST charge is simply

$$Q_{\text{DS}} = \sum_{n \in \mathbb{Z}} c_{-n} (J_n^- - \delta_{n,0}), \quad Q_{\text{DS}}^2 = 0. \quad (2.26)$$

The cylinder and plane are related by one unit of spectral flow. With spectral-flow parameter ω the flow takes the standard form

$$\begin{aligned} \tilde{J}_n^3 &= J_n^3 - \frac{k}{2} \omega \delta_{n,0}, \\ \tilde{J}_n^\pm &= J_{n \pm \omega}^\pm, \\ \tilde{L}_n^{\text{Sug}} &= L_n^{\text{Sug}} + \omega J_n^3 - \frac{k}{4} \omega^2 \delta_{n,0}. \end{aligned} \quad (2.27)$$

It is immediate that for $\omega = -1$ one has $\tilde{J}_{-1}^- = J_0^-$. Thus the Brown-Henneaux condition on the cylinder is a one-unit spectrally flowed version of the plane DS constraint. We will see that it fixes the physical Hamiltonian below.

This difference has direct representation-theoretic consequences. At level zero the cylinder reduction requires $(J_0^- - 1)|\psi\rangle = 0$. So a surviving state must have nonzero J_0^- eigenvalue. On any finite-dimensional horizontal module, J_0^- is nilpotent and therefore cannot have eigenvalue one. Equivalently, $J_0^- - 1$ is invertible, so the level-zero BRST cohomology is empty. This includes the trivial horizontal module underlying the affine vacuum. Because they are *infinite*-dimensional, the discrete and principal-series modules relevant for us, evade this: in the appropriate completion the same constraint can have nontrivial solutions, which furnish the level-zero classes of the flowed reduction.

For standard quantum DS reduction and its BRST formulation see [27, 28, 29, 30, 31, 32]. Spectrally flowed DS reduction is discussed by Dhillon [34]. Because spectral flow changes the reduction functor, we do not invoke the usual unflowed highest-weight vanishing theorems as an all-level proof for the modules used here. The flowed BRST complex, including its extension to relaxed modules, is analyzed in detail in [33]: below we retain only the ingredients needed in this paper.

2.5.2 BRST Grading, Central Charge and Physical Weight

The DS improvement is

$$T_{\text{imp}} = T_{\text{Sug}} - \partial J^3, \quad (2.28)$$

under which J^- has weight zero, as required for a nonzero constant constraint. Including the (b, c) ghosts of weights $(0, 1)$ gives

$$T_{\text{tot}} = T_{\text{Sug}} - \partial J^3 + T_{bc}. \quad (2.29)$$

Its central charge is

$$c = \frac{3k}{k-2} + 6k - 2 = 1 + 6Q^2, \quad Q = b + \frac{1}{b}, \quad b^{-2} = k - 2. \quad (2.30)$$

This is the usual Liouville parametrization and tends to the Brown-Henneaux result $c = 6k$ at large k .

The BRST complex carries a useful level grading. With Δ_j the Sugawara weight we introduced earlier, define

$$N = L_0^{\text{Sug}} + L_0^{\text{gh}} - \Delta_j. \quad (2.31)$$

Since¹⁶

$$[Q_{\text{DS}}, L_0^{\text{Sug}} + L_0^{\text{gh}}] = 0,$$

one has $[Q_{\text{DS}}, N] = 0$, and the cohomology can therefore be computed level by level.

A separate question is which Virasoro zero mode acts on this cohomology. An operator representing L_0 on the reduced theory must preserve BRST-closed states (and BRST equivalence classes), so we seek a representative commuting with Q_{DS} . The zero mode of the spectrally unflowed DS stress tensor T_{tot} is $L_0^{\text{imp}} + L_0^{\text{gh}} = L_0^{\text{Sug}} + J_0^3 + L_0^{\text{gh}}$. For the cylinder BRST charge, however, this is not BRST invariant. Indeed, using (2.26) and $[J_n^-, J_0^3] = J_n^-$,

$$[Q_{\text{DS}}, J_0^3] = Q_{\text{DS}} + c_0 \implies [Q_{\text{DS}}, L_0^{\text{imp}} + L_0^{\text{gh}}] = Q_{\text{DS}} + c_0 \neq 0. \quad (2.32)$$

Thus the unflowed improved zero mode does not in general preserve the BRST-closed subspace. This is precisely where the spectral flow of the cylinder reduction matters: the stress tensor must be taken in the same flowed frame as the current constraint. With the ghost sector unchanged, the corresponding improved total stress tensor is

$$\tilde{T}_{\text{tot}} = \tilde{T}_{\text{Sug}} - \partial \tilde{J}^3 + T_{bc},$$

¹⁶The vanishing of this commutator can be seen directly mode by mode. With respect to the Sugawara stress tensor, J^- is a weight-one current, and hence $[L_0^{\text{Sug}}, J_n^-] = -nJ_n^-$. Similarly, since c has weight one with respect to the ghost stress tensor, $[L_0^{\text{gh}}, c_n] = -nc_n$. Therefore, using that the matter and ghost sectors commute, $[L_0^{\text{Sug}} + L_0^{\text{gh}}, c_{-n}J_n^-] = (n - n)c_{-n}J_n^- = 0$. The remaining subtraction term in Q_{DS} is $-c_0$, which also commutes with $L_0^{\text{Sug}} + L_0^{\text{gh}}$. Thus every term in (2.26) has level zero, establishing the commutator above.

and we define its zero mode to be the physical Virasoro generator,

$$\begin{aligned}
L_0^{\text{phys}} &\equiv \tilde{L}_0^{\text{Sug}} + \tilde{J}_0^3 + L_0^{\text{gh}} \\
&= \left(L_0^{\text{Sug}} - J_0^3 - \frac{k}{4} \right) + \left(J_0^3 + \frac{k}{2} \right) + L_0^{\text{gh}} \\
&= L_0^{\text{Sug}} + L_0^{\text{gh}} + \frac{k}{4}.
\end{aligned} \tag{2.33}$$

In the last two lines we used the $\omega = -1$ spectral-flow relations. Since the final expression differs from $L_0^{\text{Sug}} + L_0^{\text{gh}}$ only by a constant, it satisfies

$$[Q_{\text{DS}}, L_0^{\text{phys}}] = 0,$$

and therefore defines the Virasoro zero mode on the BRST cohomology.

Note that the J_0^3 dependence cancels. We briefly explain why this is important. Before spectral flow, the improved zero mode L_0^{imp} contains an explicit J_0^3 contribution, as we saw earlier. The level-zero BRST solution, however, must necessarily be a superposition of states with different J_0^3 eigenvalues¹⁷. The rest of the terms in L_0^{imp} (the Sugawara term and the ghost piece) are fixed at level zero, within a fixed horizontal representation. So the only term whose eigenvalue varies across the different J_0^3 components is J_0^3 itself. This means that the BRST solution cannot be an eigenstate of the unflowed improved L_0^{imp} . The flowed version on the other hand does not have J_0^3 -dependence, and therefore this problem goes away. In other words: BRST forces the state away from definite J_0^3 weights, but spectral flow comes to the rescue by removing J_0^3 from the physical L_0 .

On level zero one obtains the following formula (which applies for general modules):

$$h = \frac{C_2}{k-2} + \frac{k}{4}. \tag{2.34}$$

For the discrete series this gives

$$h_j = \frac{k}{4} - \frac{j(j-1)}{k-2} = \frac{c-1}{24} - \frac{(j-\frac{1}{2})^2}{k-2} = \frac{Q^2}{4} + P_j^2, \quad P_j = \frac{i(j-\frac{1}{2})}{\sqrt{k-2}}. \tag{2.35}$$

Thus the elliptic branch has imaginary Liouville momentum and lies below the threshold $(c-1)/24$. On the branch¹⁸ $j > \frac{1}{2}$, the interval¹⁹

$$\frac{1}{2} < j < \frac{k}{2} \tag{2.36}$$

¹⁷Note that the BRST constraint $J_0^- |\psi\rangle = |\psi\rangle$ is incompatible with a definite J_0^3 eigenvalue.

¹⁸The reflected branch $j < \frac{1}{2}$ is distinct at the unreduced $SL(2, \mathbb{R})$ level, but corresponds to $\rho = j - \frac{1}{2} < 0$. The associated metric geometry and reduced Virasoro data are insensitive to $\rho \rightarrow -\rho$. We therefore choose the branch $\rho > 0$, equivalently $j > \frac{1}{2}$.

¹⁹An interesting comparison is with the Maldacena–Ooguri window $\frac{1}{2} < j < \frac{k-1}{2}$ [53]. There also, the no-ghost bound itself is $j < k/2$. The stronger upper bound follows from requiring the worldsheet WZW

corresponds to $0 < h_j < (c-1)/24$. The upper endpoint is particularly instructive:

$$j = \frac{k}{2} \implies h_j = 0, \quad P_j = \frac{i}{2} \left(b + \frac{1}{b} \right) = \frac{iQ}{2}. \quad (2.37)$$

This is the $(1,1)$ degenerate Virasoro weight. It is important, however, to distinguish the value of the highest weight from the Virasoro module itself. The flowed DS reduction of the $j = k/2$ affine module gives us the full $h = 0$ Verma module. It does not by itself implement the level-one quotient that turns this into the Virasoro vacuum module. But for the cylinder reduction done here, we believe it is quite natural to do this quotient.

Firstly, it is natural that the $j = k/2$ sector (rather than say, the affine vacuum module ($j = 0$)) is what corresponds to the Virasoro vacuum. This is what the bulk geometry requires: smooth global AdS_3 has central spatial holonomy, invisible in the metric – in the fundamental representation (2.7) it is $-\mathbf{1}$ – and in the semiclassical dictionary of Section 2.4.3 it is the $j = k/2$ sector that carries it.²⁰ This is consistent also with the fact that the vacuum has extra isometries which need quotienting, as reflected in the orbit language: it corresponds to the exceptional orbit. Thus the absence of the Wilson line should not be identified with the gravitational vacuum in this reduction, and there is no contradiction in the affine vacuum module having trivial BRST cohomology.

There is also a natural “upstairs” motivation for the additional quotient. Precisely at $j = k/2$ the affine module develops a singular sub-module. The flowed BRST reduction is nevertheless blind to this reducibility: it still produces the full $h = 0$ Virasoro Verma module [33]. But if one views the quotiented module upstairs as the more fundamental object, it is natural to quotient by the singular sub-module downstairs as well. And there is indeed such a candidate at $h = 0$: the level-one singular sub-module generated by $L_{-1}|0\rangle$. The result is the Virasoro vacuum module. Thus the vacuum is obtained by first reaching the $h = 0$ endpoint of the generic family and only then performing the extra quotient.

This story-arc gives further support for the “regulate-and-quotient” prescription of [2], in which the vacuum is treated as a limiting normal orbit and the quotient associated with its enhanced stabilizer is imposed only afterwards. In particular, the normal-orbit quantization

spectrum to be closed under spectral flow, since one unit of flow maps $\widehat{\mathcal{D}}_j^+$ to $\widehat{\mathcal{D}}_{k/2-j}^-$, together with the $j > 1/2$ lower bound from the zero-mode harmonic analysis. That additional requirement is not imposed here: spectral flow here is part of the reduction defining the gravitational state space. Thus the Maldacena–Ooguri tightening of the upper bound does not directly apply. The endpoint $j = k/2$ is instead singled out by $h_j = 0$, which eventually becomes the Virasoro vacuum.

²⁰Combining the exact relation $\rho = j - \frac{1}{2}$ with the classical identification $\alpha = 2\rho/k$ gives $\alpha = 1 - 1/k$ at $j = k/2$ rather than $\alpha = 1$ on the nose. This residual $O(1/k)$ is below the resolution of the semiclassical matching (2.20). The exact finite- k statement is that the flowed DS reduction reaches $h = 0$ (equivalently, $P = iQ/2$) at $j = k/2$. The vacuum module requires the additional level-one quotient.

gives a single central charge $c = C + 1$ across the sectors, where C is the classical central charge of the orbit. By (2.37),

$$P^2|_{j=k/2} = -\frac{Q^2}{4} = -\frac{c-1}{24},$$

while the exceptional vacuum orbit has $2\pi b_0 = -C/24$. Using the normal-orbit relation $2\pi b_0 = h - (c-1)/24$, the two agree precisely when $C = c - 1$. This should be contrasted with the $c = C + 13$ result obtained by quantizing the exceptional orbit as an independent object [26, 54]. The correspondence is therefore quite direct: generic j in the interval (2.36) reduces to a Virasoro Verma module, $j = k/2$ reaches its $h = 0$ endpoint, and the subsequent level-one quotient produces the vacuum module. The singular sub-module that appears simultaneously in the $j = k/2$ affine module provides an independent indication that precisely this endpoint requires special quotienting [33].

This also gives a sharp check on the cylinder reduction: applying the unflowed plane improvement to the original lowest-weight vector would instead give

$$h_{\text{pl}} = \Delta_j + j = \frac{j(k-1-j)}{k-2}, \quad h_{\text{pl}}|_{j=k/2} = \frac{k}{4}, \quad (2.38)$$

which does not reproduce the vacuum endpoint. Finally, inserting the semiclassical identification (2.20) into (2.35) gives

$$h_j = \frac{k}{4} - \frac{j^2}{k} + O(1) = \frac{c}{24}(1 - \alpha^2) + O(1), \quad (2.39)$$

in agreement with (2.21). The gravitational weight is therefore an output of the flowed Brown-Henneaux reduction.

2.5.3 A Single Primary and the Virasoro Tower

It remains to see what happens to the infinite horizontal multiplet. At level zero and ghost number zero, the BRST condition (2.26) is simply

$$(J_0^- - 1)|\psi\rangle = 0. \quad (2.40)$$

For the discrete series this equation has an honest solution in the Hilbert-space completion of the horizontal module. Writing

$$|\psi\rangle = \sum_{p \geq 0} c_p^W (J_0^+)^p |j, j\rangle, \quad (2.41)$$

and using

$$J_0^- (J_0^+)^p |j, j\rangle = p(2j + p - 1) (J_0^+)^{p-1} |j, j\rangle, \quad (2.42)$$

one finds

$$(p+1)(2j+p)c_{p+1}^W = c_p^W, \quad c_p^W = \frac{c_0^W}{p!(2j)_p}. \quad (2.43)$$

For $j > 0$ the denominator never vanishes, so the solution is unique up to its overall normalization. Moreover, with the standard unitary discrete-series norm, $\|(J_0^+)^p|j, j\rangle\|^2 = p!(2j)_p$, and hence

$$\langle\psi|\psi\rangle = |c_0^W|^2 \sum_{p=0}^{\infty} \frac{1}{p!(2j)_p} = |c_0^W|^2 {}_0F_1(; 2j; 1) < \infty. \quad (2.44)$$

Thus the nonzero cylinder constraint selects a single level-zero state. In the discrete series, it is in addition, normalizable. We use the ghost vacuum convention $b_{n \geq 0}|0\rangle_{\text{gh}} = 0$ and $c_{n \geq 1}|0\rangle_{\text{gh}} = 0$, so the level-zero ghost sector is spanned by $|0\rangle_{\text{gh}}$ and $c_0|0\rangle_{\text{gh}}$ and contains no ghost-number -1 state. This closed state therefore supplies the unique ghost-number-zero BRST class. It is the Virasoro primary of weight (2.35). The principal-series analogue in the next section is instead represented by a formal, non-normalizable Whittaker vector in the appropriate completion. The barred zero-mode argument is identical in the barred basis, with $J^\pm$ replaced by $\bar{J}^\pm$.

At positive level the reduction is explicitly cohomological: the constraint and its gauge symmetry are expected to reorganize the three affine-current towers into the single Virasoro tower generated by the reduced stress tensor. For the *flowed* reduction used here, the all-level statement is not being inferred directly from the standard unflowed highest-weight vanishing theorems. The detailed BRST analysis will be presented in [33]. The all-level statement used in the remainder of this paper for generic j is

$$H_{\text{DS,flow}}^0(\widehat{\mathcal{D}}_j^+) \cong \mathcal{V}_{h_j}. \quad (2.45)$$

What has been established directly above is the level-zero part of this statement and the physical grading and weight. We have checked the cohomology explicitly at low levels and various ghost numbers, but a full discussion (for both discrete and continuous series) is part of [33]. With that reduction understood, the physical descendants are

$$L_{-n_1} \cdots L_{-n_p} \bar{L}_{-m_1} \cdots \bar{L}_{-m_q} |h_j, \bar{h}_j\rangle, \quad n_i, m_i > 0. \quad (2.46)$$

Their levels are measured by the BRST-invariant grading (2.31).

2.6 Summary and a Look Ahead

Let us summarize what we have done. An elliptic Wilson line prepares the unreduced puncture sector $\widehat{\mathcal{D}}_j^+ \otimes \widehat{\mathcal{D}}_{\bar{j}}^+$ on the spatial disc. Its holonomy fixes the semiclassical defect parameter through $j = k\alpha/2 + O(1)$. Brown-Henneaux boundary conditions then impose a

nonzero constant current on the boundary cylinder, which we normalize to $J_0^- = 1$. Relative to the common plane convention this is a one-unit spectrally flowed DS reduction. The flow fixes the physical Hamiltonian and gives

$$L_0^{\text{phys}} = L_0^{\text{Sug}} + L_0^{\text{gh}} + \frac{k}{4}, \quad h = \frac{C_2}{k-2} + \frac{k}{4}. \quad (2.47)$$

For $C_2 = -j(j-1)$ this is the exact sub-threshold weight (2.35): the level-zero constraint has a unique normalizable solution in the completed discrete-series module, and $j = k/2$ leads to the Virasoro vacuum.

The same Brown-Henneaux reduction will be used in the hyperbolic sector. What changes there is the horizontal representation: the principal continuous series is relaxed and has no extremal vector. The module-independent part of the calculation above nevertheless continues immediately. In particular, for $C_2 = s^2 + \frac{1}{4}$ one obtains

$$h_P = \frac{c-1}{24} + P^2, \quad P = \frac{s}{\sqrt{k-2}}. \quad (2.48)$$

The existence and uniqueness of the level-zero class can again be shown directly, while the all-level cohomology of the relaxed module requires additional care. We will use the result in the next section and defer the detailed BRST analysis to [33].

3 BTZ Microstates and Super-Threshold Primaries

We now turn to the main construction of this paper: the bulk state associated with a Virasoro primary above the BTZ threshold. The logic is the same as in the elliptic sector of Section 2, with one important change upstairs. The Wilson line now carries a principal continuous-series representation, appropriate to hyperbolic holonomy, so the unreduced puncture sector is a relaxed affine module. Brown-Henneaux boundary conditions then impose the same cylinder reduction derived in Section 2.5: the nonzero constant constraint is $J_0^- = 1$, and relative to the plane constraint this is a one-unit spectrally flowed Drinfel'd-Sokolov reduction.

Thus the quantum construction has the schematic form

$$\text{principal-series Wilson line} \longrightarrow \widehat{\mathcal{C}}_s^0 \xrightarrow{H_{\text{DS,flow}}^0} \mathcal{V}_{h_P}. \quad (3.1)$$

The Wilson line fixes the hyperbolic conjugacy class of the spatial holonomy. The flowed reduction removes the unreduced horizontal zero-mode label and produces the Virasoro primary together with its boundary-graviton descendants. The module-independent part of this reduction, including the physical Hamiltonian and central charge, was already derived in Section 2.5. Here we only need to explain how it acts on the principal continuous series.

The resulting primary has real Liouville momentum P and weight

$$h_P = \frac{c-1}{24} + P^2, \quad (3.2)$$

and therefore lies above the threshold $(c-1)/24$. Distinct heavy primaries, rather than their universal Virasoro descendants, are the microscopic labels relevant to the black-hole entropy. The descendants are subleading boundary-graviton excitations built on each such primary [2].

3.1 BTZ Geometry and Hyperbolic Holonomy

For illustration, consider first the non-rotating BTZ geometry [55] in Euclidean signature,

$$ds^2 = (r^2 - r_+^2) d\tau^2 + \frac{dr^2}{r^2 - r_+^2} + r^2 d\phi^2, \quad \phi \sim \phi + 2\pi. \quad (3.3)$$

When τ is periodically identified with inverse temperature $\beta = 2\pi/r_+$, this is the smooth Euclidean BTZ saddle. In the present section we use the metric only to read off the classical holonomy associated with a heavy primary sector [2]. Without the thermal identification, the geometry will be singular at the horizon. We will discuss this in detail, later.

The ADM mass is

$$M = \frac{r_+^2}{8G_N}, \quad (3.4)$$

and the classical Brown-Henneaux weights are

$$h_L = h_R = \frac{c}{24} (1 + r_+^2). \quad (3.5)$$

At finite k the exact reduced Virasoro weight will instead be written as $h_P = (c-1)/24 + P^2$. The difference between $c/24$ and $(c-1)/24$ is an order-one quantum shift and is subleading in the semiclassical large- c regime.

In Chern-Simons language the gauge-invariant datum carried by the Wilson line is the conjugacy class of the holonomy around a small spatial circle linking it. In the BTZ sector this holonomy is hyperbolic. For one chiral copy we write

$$\text{Hol}(A) \sim \begin{pmatrix} e^{\pi\lambda} & 0 \\ 0 & e^{-\pi\lambda} \end{pmatrix}, \quad \lambda > 0. \quad (3.6)$$

For the non-rotating geometry $\lambda = r_+$. The corresponding Wilson-line orbit is therefore hyperbolic. A smooth black hole would correspond to the demand that the thermal holonomy is trivial, in the Chern-Simons language.

3.2 Principal Continuous Series and the Flowed Reduction

The horizontal representation appropriate to a hyperbolic orbit is the principal continuous series of $SL(2, \mathbb{R})$. We denote it by $\mathcal{C}_s^\epsilon$, with $s > 0$ and J_0^3 spectrum $m \in \epsilon + \mathbb{Z}$. The global label ϵ will play no role below²¹, and we fix $\epsilon = 0$. With the Casimir convention of Section 2.4.1,

$$C_2(\mathcal{C}_s^\epsilon) = s^2 + \frac{1}{4}. \quad (3.7)$$

Equivalently one may write $j = \frac{1}{2} + is$, but the invariant label s will be more useful.

There is no highest- or lowest-weight vector in $\mathcal{C}_s^0$. The unreduced affine module is therefore the relaxed module induced from the full horizontal representation,

$$\widehat{\mathcal{C}}_s^0 = \text{span} \{ J_{-n_1}^{a_1} \cdots J_{-n_p}^{a_p} v ; n_i > 0, v \in \mathcal{C}_s^0 \}. \quad (3.8)$$

This is the natural sector produced by the Wilson line before Brown-Henneaux reduction. In particular, its infinite horizontal multiplet is unreduced current-algebra data.

The gravitational reduction is exactly the cylinder reduction already derived in Section 2.5.1. Classically the corresponding Drinfel'd-Sokolov form of the boundary connection is

$$a_\phi = F - \frac{2\pi}{k} \mathcal{L}(\phi) E, \quad (3.9)$$

while quantum mechanically the constraints are $J_n^- = \delta_{n,0}$. Because the current has weight one before the DS improvement, a constant on the boundary cylinder fixes J_0^- , whereas a constant in the standard plane presentation fixes J_{-1}^- . As shown in Section 2.5.1, the two are related by spectral flow with $\omega = -1$. The corresponding flowed BRST complex is developed in [33]. A noteworthy fact is that the level-by-level cohomology calculations in discrete and continuous series are largely parallel (summarized at the end of Appendix B, details in [33]), and differ only in the value of the Casimir.

For the present paper, the crucial module-independent result was already obtained in eq. (2.34):

$$h = \frac{C_2}{k-2} + \frac{k}{4}. \quad (3.10)$$

Substituting (3.7) gives

$$P = \frac{s}{\sqrt{k-2}}, \quad (3.11)$$

²¹Note however that it *is* part of the Wilson line label. It however does not enter the holonomy conjugacy class, and the reduction below is carried out at fixed ϵ and returns the same Virasoro momentum P and an isomorphic Virasoro module for every value of ϵ . We will view this as the statement that AdS_3 gravity is sensitive only to the reduced labels: it is a Virasoro theory, not an $SL(2, \mathbb{R})$ theory. But there may be some physics here that is of interest, and worth coming back to.

and hence²²

$$h_P = \frac{k}{4} + \frac{s^2 + \frac{1}{4}}{k-2} = \frac{Q^2}{4} + P^2 = \frac{c-1}{24} + P^2, \quad Q = b + \frac{1}{b}, \quad b^{-2} = k-2. \quad (3.12)$$

Thus the super-threshold weight is a direct output of the flowed Brown-Henneaux reduction. No J_0^3 eigenvalue appears: the J_0^3 dependence cancels from the physical Hamiltonian under the spectral flow, exactly as in Section 2.5.

What remains module-dependent is the emergence of the primary class and the structure of its positive-level cohomology. At level zero the answer is elementary. In the natural completion of the principal series, the nonzero-eigenvalue condition

$$(J_0^- - 1)|\psi\rangle = 0 \quad (3.13)$$

has a one-dimensional space of solutions for fixed s : J_0^- shifts the two-sided J_0^3 lattice with a coefficient that never vanishes for $s > 0$. Appendix B presents the short discussion. Thus no individual horizontal label m survives as a label of the reduced primary. At positive affine level the statement is cohomological. The all-level statement we expect is

$$H_{\text{DS,flow}}^0(\widehat{\mathcal{C}}_s^0) \cong \mathcal{V}_{h_P} \quad (3.14)$$

for generic real P . As in the discrete sector, this is a reorganization of the affine module by cohomology [33].

3.3 Matching to BTZ Parameters

For the hyperbolic orbit, take $H_{\text{hyp}} = \frac{1}{2}\sigma_3$ and $Q_{\text{hyp}}^{\text{orb}} = 2\rho_{\text{hyp}}H_{\text{hyp}}$. Then eq. (2.10) gives holonomy eigenvalues $e^{\pm 2\pi\rho_{\text{hyp}}/k}$. Standard coadjoint-orbit quantization gives the exact continuous-series dictionary $\rho_{\text{hyp}} = s$ [52]. This fits naturally with the elliptic result $\rho = j - \frac{1}{2}$. At the level of the complexified orbit radius, the continuation $\rho^2 \rightarrow -\rho_{\text{hyp}}^2$ (equivalently $\rho \rightarrow \pm i\rho_{\text{hyp}}$) takes the elliptic holonomy conjugacy class into the hyperbolic one, and correspondingly

$$C_2 = \frac{1}{4} - \rho^2 \longrightarrow \frac{1}{4} + \rho_{\text{hyp}}^2 = \frac{1}{4} + s^2,$$

in agreement with the principal-series Casimir (3.7). Comparing the hyperbolic orbit holonomy with the classical BTZ holonomy (3.6) then gives $\rho_{\text{hyp}} = k\lambda/2$ semiclassically. This means that

$$s = \frac{k\lambda}{2} + O(1), \quad \lambda = \frac{2s}{k} + O(1/k). \quad (3.15)$$

For the non-rotating BTZ geometry, $\lambda = r_+$, and therefore

$$M = \frac{r_+^2}{8G_N} = \frac{s^2}{2k^2G_N} (1 + O(1/k)). \quad (3.16)$$

²²These expressions have also been presented in [56].

Using $c = 6k$ at leading semiclassical order, the Brown-Henneaux weight is

$$h = \frac{c}{24} (1 + \lambda^2) = \frac{c}{24} + \frac{s^2}{k} + O(1). \quad (3.17)$$

This is precisely the large- k limit of the exact reduced expression (3.12).

For a rotating BTZ sector the two chiral Wilson lines carry independent labels s_L, s_R , or equivalently P_L, P_R , with

$$L_0 - \frac{c}{24} = \frac{\ell M + J}{2}, \quad \bar{L}_0 - \frac{c}{24} = \frac{\ell M - J}{2}, \quad (3.18)$$

and semiclassically

$$s_L = \frac{k\lambda_L}{2} + O(1), \quad s_R = \frac{k\lambda_R}{2} + O(1), \quad \lambda_L = \frac{r_+ + r_-}{\ell}, \quad \lambda_R = \frac{r_+ - r_-}{\ell}. \quad (3.19)$$

The exact reduced weights are $h_L = (c - 1)/24 + P_L^2$ and $h_R = (c - 1)/24 + P_R^2$, with $P_{L,R} = s_{L,R}/\sqrt{k - 2}$.

3.4 The Path-Integral State

Before reduction, the punctured-disc state space in the two chiral sectors is

$$\mathcal{H}_{\text{aff}}(D^2; s_L, s_R) = \widehat{\mathcal{C}}_{s_L}^0 \otimes \widehat{\mathcal{C}}_{s_R}^0. \quad (3.20)$$

The barred principal-series module uses the same opposite-Borel convention fixed in Section 2.4.2. The Euclidean path integral on the half-cylinder $M_- = D^2 \times (-\infty, 0]$, with principal-series Wilson lines ending on the puncture of the final slice, prepares an unreduced affine state schematically of the form

$$|\Psi_{s_L, s_R}\rangle_{\text{aff}} = \int_{\mathcal{P}_{\text{aff}}} [\mathcal{D}A] [\mathcal{D}\tilde{A}] W_{C_{s_L}^0}[A] W_{\bar{C}_{s_R}^0}[\tilde{A}] e^{-S_E[A, \tilde{A}]} |A, \tilde{A}\rangle_{\tau=0}. \quad (3.21)$$

As in Section 2.4.2, the open Wilson-line endpoint is representation-valued. Here there is no distinguished extremal vector in the horizontal principal series, so the half-cylinder does not canonically select a particular J_0^3 basis vector. This causes no ambiguity in the Brown-Henneaux theory: the entire affine sector is the input to the flowed DS reduction, and the one-dimensional level-zero cohomology supplies the Virasoro primary.

After imposing the Brown-Henneaux constraint in both chiralities, the reduction acts on the affine puncture sector,

$$H_{\text{DS,flow}}^0(\mathcal{H}_{\text{aff}}(D^2; s_L, s_R)) \cong \mathcal{V}_{h_L} \otimes \bar{\mathcal{V}}_{h_R} = \mathcal{H}_{\text{phys}}(D^2; P_L, P_R). \quad (3.22)$$

The Wilson-line half-cylinder supplies the unreduced puncture sector entering this BRST complex, while imposing the Brown-Henneaux reduction prepares the physical cohomology

class. At level zero this is the primary $|P_L, P_R\rangle_{\text{phys}} = |h_L\rangle_L \otimes |h_R\rangle_R$. For $P_L = P_R = P$ it is the spinless heavy primary $|h_P, h_P\rangle$. This is the precise sense in which the principal-series Wilson line constructs a super-threshold Virasoro primary: the Wilson line fixes the unreduced orbit and holonomy, while the physical primary is selected by the flowed Brown-Henneaux reduction.

3.5 Virasoro Descendants

The reduced primary carries the usual Virasoro descendant tower,

$$\mathcal{V}_{h_L} \otimes \bar{\mathcal{V}}_{h_R} = \text{span} \left\{ L_{-n_1} \cdots L_{-n_p} \bar{L}_{-\bar{n}_1} \cdots \bar{L}_{-\bar{n}_q} |h_L, h_R\rangle ; n_i, \bar{n}_i > 0 \right\}. \quad (3.23)$$

These are boundary-graviton excitations of a fixed primary sector. A descendant at levels $(N, \bar{N})$ has $L_0 = h_L + N$ and $\bar{L}_0 = h_R + \bar{N}$, so the Wilson line fixes the primary holonomy data rather than (the exact energy of) every descendant. The latter are controlled by the non-zero-mode Virasoro generators, which act only on the boundary: they are boundary hair. The BRST-invariant level grading is the same one introduced in (2.31).

The super-threshold construction may therefore be summarized as

$$\boxed{\mathcal{C}_s^0 \longrightarrow \hat{\mathcal{C}}_s^0 \xrightarrow{H_{\text{DS,flow}}^0} \mathcal{V}_{h_P}, \quad h_P = \frac{c-1}{24} + P^2, \quad P = \frac{s}{\sqrt{k-2}}.} \quad (3.24)$$

The subsequent sections study the semiclassical saddle associated with this individual heavy primary and then its relation to the smooth thermal black hole.

4 The Semi-Classical Microstate: A Singular Horizon

In Sections 2 and 3, we constructed the quantum state $|P\rangle$ as a non-perturbative object in Chern-Simons gravity by starting from the Euclidean half-cylinder path integral with a continuous-series Wilson line at the origin of the disc and imposing the Brown-Henneaux reduction. In this section, we study what this state looks like in the large- k limit, which we will view as a type of (semi-)classicality²³. Throughout this section, relations between the exact representation label and geometric quantities are understood at leading large k , and the subleading corrections in (3.15) are suppressed.

The result is as follows. In the large- k WKB limit, the path integral (3.21) is dominated by a saddle point. The saddle-point connection is determined by three ingredients – chirality demand on the gauge field, the holonomy fixed by the representation label s , and the

²³But note the final bullet point in the Introduction.

Drinfel'd-Sokolov²⁴ boundary conditions. Translating from the gauge-field language to the metric language via the dictionary (2.1), the saddle is the BTZ metric (3.3) with horizon radius $r_+ = \lambda = 2s/k$. However the microstate saddle has a fundamentally different global structure from the thermal Euclidean BTZ²⁵. The Euclidean time direction is *non-compact*, there is no periodicity condition, and the geometry terminates at the horizon with a Wilson line source. In the metric language, the horizon has a finite perimeter $2\pi r_+$, but it is naturally viewed as a singular (geodesically incomplete) geometry with a source. If one lets this microstate run in a thermal loop for some fixed external temperature β , there is a codimension-2 conical excess/defect singularity at the horizon location, with smoothness only at the fine-tuned value $\beta = 2\pi/r_+$. We emphasize that these are all semi-classical statements at large- k and the state is well-defined as a quantum state at finite- k for any β .

In Lorentzian signature, this singularity persists: the semi-classical geometry of the individual microstate is the exterior BTZ metric with a singular horizon that has no natural extension into a smooth Kruskal diagram. The smooth horizon is recovered only as the property of an ensemble of states, as we will see in this and the next section.

These observations were anticipated in [2] (see especially Section 4.2 of that reference), where the relevant Euclidean geometry was described in the Chern-Simons framework. The present section provides the explicit saddle-point statement in the context of the path-integral construction of the microstate. The solutions are simple and live in the vicinity of well-known expressions. But we will be deliberately slow, because the boundary/regularity conditions are non-standard and we wish to be fully aware of assumptions, when we are making them.

4.1 The WKB Limit

The microstate is defined by the bulk Chern-Simons path integral (3.21) on the half-cylinder $M_- = D^2 \times (-\infty, 0]$, with a continuous-series Wilson line along the time axis at the centre of the disc. The integration variables are the bulk gauge fields A and $\tilde{A}$, and it is this object that we analyse at large k .

Since the action (2.3) carries an overall factor of the level k , the integral is controlled by

²⁴We will view chirality together with Drinfel'd-Sokolov gauge, as the definition of Brown-Henneaux boundary conditions. Note that our construction canonicalizes this notion of “asymptotically AdS_3 ”.

²⁵The discussions in this section straightforwardly generalize to rotating BTZ as well, but we restrict to the non-rotating case for clarity.

a stationary-phase approximation²⁶. With $(A_*, \tilde{A}_*)$ the dominant stationary configuration,

$$|\Psi_{s_L, s_R}\rangle_{\text{aff}} \sim e^{-S_E[A_*, \tilde{A}_*]} |A_*, \tilde{A}_*\rangle_{\tau=0} \left(1 + \mathcal{O}(1/k)\right), \quad (4.1)$$

and the classical geometry of the microstate is the one built from $(A_*, \tilde{A}_*)$. The saddle solves the Chern-Simons equations of motion on M_- in the presence of the line source, subject to:

- **Wilson line source and holonomy:** by (2.9) the field strength of each connection has a delta function on the line and vanishes away from it. The connection is flat on the punctured disc, and the conjugacy class of the holonomy linking the puncture is fixed by the Wilson-line label: for the principal continuous series it is hyperbolic with boost parameter $\lambda = 2s/k$, by (3.15).
- **Boundary conditions:** the Brown-Henneaux conditions are imposed at the asymptotic boundary. We take these to be chirality of the two connections together with the Drinfel'd-Sokolov gauge (3.9).
- **Global structure:** the Euclidean time direction runs over the non-compact range $(-\infty, 0]$. No periodic identification is imposed, because none is part of the definition of (3.21).

Two comments. First, the path integral (3.21) does not select a particular vector in the principal-series zero-mode multiplet, but the saddle sees the Wilson line only through its coadjoint orbit, and hence only through the Casimir and the holonomy class. Second, the delta function above is one in the Chern-Simons field strength, and we will often refer to it as the delta-function singularity in the curvature. Its metric translation depends on the global choices we make, and will also be discussed. Note that the singularity in the metric language is also technically a curvature singularity. But it is not a curvature divergence singularity as in Schwarzschild, it is a delta function defect type singularity.

4.2 The Saddle Connection

We work in Lorentzian signature in this subsection and continue to Euclidean signature at the end using $t = -i\tau$.

²⁶One might worry about the uniqueness of the saddle. The only gauge-invariant data (or allowed gauge-variant data, which can only happen at boundary) are the holonomy controlled by the source (and the boundary gravitons). This is a standard fact about semi-classical bulk, and our statements will be precisely compatible with that. We will ignore boundary gravitons, they can be added without any change in our horizon/singularity discussions.

Fefferman–Graham gauge. We adopt the radial gauge

$$A_\rho = H, \quad \tilde{A}_\rho = -H, \quad (4.2)$$

where ρ is the radial coordinate on the disc, increasing from the puncture out to the asymptotic boundary at $\rho \rightarrow \infty$. Flatness in the radial direction, $\partial_\rho A_i + [H, A_i] = 0$ for $i = \phi, t$, fixes the ρ -dependence of the remaining components in terms of a ρ -independent reduced connection a_i ,

$$A_i(\rho) = e^{-\rho \text{ad}_H} a_i = (a_i)_E e^{-\rho} E + (a_i)_H H + (a_i)_F e^{+\rho} F, \quad (4.3)$$

where we used $[H, E] = E$ and $[H, F] = -F$, and similarly for $\tilde{A}_i$ with $\rho \rightarrow -\rho$. The Drinfel'd-Sokolov form (3.9) of the Brown-Henneaux conditions removes the H component and normalises the leading term of a_ϕ to unity in the F direction [26],

$$a_\phi = -\frac{2\pi}{k} \mathcal{L} E + F, \quad (4.4)$$

with the constant $\mathcal{L}$, free. For the second connection the same conditions apply with E and F interchanged, reflecting the opposite chirality, so that $\tilde{a}_\phi = -\frac{2\pi}{k} \tilde{\mathcal{L}} F + E$. This is precisely the barred convention of Section 2.4.2: $\bar{J}^- = E$ is the fixed DS component, while $\bar{J}^+ = F$ carries $\tilde{\mathcal{L}}$.

Holonomy constraint. The holonomy around the ϕ -circle at any fixed ρ is conjugate to $\exp(2\pi a_\phi)$, independently of ρ . In the fundamental representation, using the choices fixed in Section 2.1, the matrix a_ϕ in (4.4) has eigenvalues $\pm \sqrt{\frac{2\pi}{k} \mathcal{L}}$. For the holonomy to be hyperbolic with boost parameter λ , i.e. to reproduce (3.6), these eigenvalues must be real and equal to $\pm \lambda/2$. This requires $\frac{2\pi}{k} \mathcal{L} > 0$ and

$$\frac{2\pi}{k} \mathcal{L} = \frac{\lambda^2}{4} = \frac{r_+^2}{4}, \quad (4.5)$$

using $\lambda = r_+$ for the non-rotating case. The same argument applied to $\tilde{a}_\phi$ gives $\frac{2\pi}{k} \tilde{\mathcal{L}} = r_+^2/4$. Therefore

$$a_\phi = -\frac{r_+^2}{4} E + F, \quad \tilde{a}_\phi = -\frac{r_+^2}{4} F + E. \quad (4.6)$$

Chirality fixes the temporal connection. The Brown-Henneaux conditions in this chiral DS form set the opposite lightcone components to zero, $a_- = 0$ and $\tilde{a}_+ = 0$. Flatness then gives $\partial_- a_+ = 0$ and $\partial_+ \tilde{a}_- = 0$, so that $a = a_+(w^+) dw^+$ and $\tilde{a} = \tilde{a}_-(w^-) dw^-$, with $w^\pm = \phi \pm t$. Consequently, $a_t = a_\phi$ and $\tilde{a}_t = -\tilde{a}_\phi$: the temporal connection is not an independent parameter. The complete saddle is therefore

$$A = H d\rho + \left(-\frac{r_+^2}{4} e^{-\rho} E + e^\rho F \right) (d\phi + dt), \quad (4.7)$$

$$\tilde{A} = -H d\rho + \left(-\frac{r_+^2}{4} e^{-\rho} F + e^\rho E\right)(d\phi - dt), \quad (4.8)$$

with $A_\tau = -iA_\phi$ and $\tilde{A}_\tau = +i\tilde{A}_\phi$ in Euclidean signature. This is the complete saddle-point connection for the microstate $|P\rangle$, with no adjustable parameters: the holonomy fixes r_+ , the boundary conditions fix the solution, and chirality fixes the temporal components.

4.3 The Metric and the Euclidean Horizon

We now translate the saddle connections (4.7)–(4.8) into the metric language using the Chern-Simons / gravity dictionary (2.1):

$$e^a J_a = \frac{\ell}{2}(A - \tilde{A}), \quad \omega^a J_a = \frac{1}{2}(A + \tilde{A}), \quad (4.9)$$

with $\ell = 1$ and $g_{\mu\nu} = \eta_{ab} e_\mu^a e_\nu^b$. The dreibein, written as $e_\mu = e_\mu^a J_a$, is

$$e_\rho = H, \quad e_\phi = \frac{f_+}{2}(F - E), \quad e_t = \frac{f_-}{2}(F + E), \quad (4.10)$$

where $f_\pm = e^\rho \pm \frac{r_+^2}{4} e^{-\rho}$. This gives $g_{\rho\rho} = 1$, $g_{\phi\phi} = f_+^2$ and $g_{tt} = -f_-^2$, with no cross terms. The radial gauge (4.2) is preserved by constant shifts of ρ , which rescale the E, F components of a_i and act as a constant Weyl rescaling of the boundary metric, leaving both the holonomy and the Drinfel'd-Sokolov conditions untouched. The shift $\rho \rightarrow \rho + \log(r_+/2)$ places the centre of the disc, where the Wilson line sits, at $\rho = 0$, and turns f_+ and f_- into $r_+ \cosh \rho$ and $r_+ \sinh \rho$. In terms of the standard BTZ radial coordinate²⁷

$$r = r_+ \cosh \rho, \quad (4.11)$$

the saddle is the Lorentzian BTZ metric

$$ds^2 = -r_+^2 \sinh^2 \rho dt^2 + d\rho^2 + r_+^2 \cosh^2 \rho d\phi^2, \quad (4.12)$$

equivalently $ds^2 = -(r^2 - r_+^2) dt^2 + dr^2/(r^2 - r_+^2) + r^2 d\phi^2$, which matches (3.3) upon Wick rotation $t \rightarrow -i\tau$. For future use, we also write the Euclidean metric in the form

$$ds_E^2 = r_+^2 \sinh^2 \rho d\tau^2 + d\rho^2 + r_+^2 \cosh^2 \rho d\phi^2. \quad (4.13)$$

²⁷The shifted coordinate ρ covers $\rho \in [0, \infty)$, corresponding to $r \in [r_+, \infty)$. In particular $\rho = 0$ corresponds to $r = r_+$. The center of the spatial disc in the Chern-Simons formulation is the horizon in the metric language. In fact, this statement is true also for the rotating BTZ black hole. This is one reason why the discussion in this section can be generalized essentially trivially to that case as well. Note also that the horizon size is finite (“size” is a metric notion!), even though the source Wilson line is sitting at the origin in these coordinates.

That the saddle is locally the BTZ metric is, of course, unsurprising: the connection is flat with hyperbolic holonomy of boost parameter $\lambda = r_+$, and since three-dimensional gravity has no local degrees of freedom, the local geometry is completely determined by the holonomy. What the local analysis does not determine, and what the path integral (3.21) does, is the global structure.

It is worth fixing the language of singularities here, since the gauge field and the metric do not use the word in the same way. In Chern-Simons variables the statement is unambiguous: by (2.9) the field strength has a delta function on the Wilson line and vanishes elsewhere. The metric translation depends on the global structure. If the Euclidean time direction is compact with period β , then $\rho = 0$ is a codimension-2 locus at which the τ -circle closes off, and unless β is tuned to $\frac{2\pi}{r_+}$, there is a conical defect or excess there, with a delta function in the curvature supported on it. This is not what is usually called a curvature singularity in the metric language, a phrase reserved for diverging curvature scalars: here the geometry is locally AdS_3 and its curvature scalars are finite everywhere except at the origin. If instead the Euclidean time direction is non-compact, as it is for the state prepared by (3.21), then $\rho = 0$ cannot be added as a regular point at all, and the invariant statement is simply that the geometry is geodesically incomplete. Therefore the context will clarify the meaning of the phrase, “singularity in curvature”. With this in mind, we now examine the geometry near $\rho = 0$.

4.3.1 The Near-Horizon Geometry

Near $\rho = 0$, the Euclidean metric (4.13) reduces to

$$ds_E^2 \approx r_+^2 \rho^2 d\tau^2 + d\rho^2 + r_+^2 d\phi^2. \quad (4.14)$$

Two features are immediately apparent:

- The ϕ -circle retains a *finite* circumference $2\pi r_+$ at $\rho = 0$. This is the *horizon perimeter*.
- The (ρ, τ) -part of the metric, $d\rho^2 + r_+^2 \rho^2 d\tau^2$, is a two-dimensional metric in “polar-like” coordinates, with ρ playing the role of the radial direction and $r_+ \tau$ playing the role of the angle.

The nature of the geometry at $\rho = 0$ depends on what range τ takes. This is where the *microstate* and the black hole (thermal state) part ways.

4.3.2 Thermal State vs. Microstate

The thermal (smooth) saddle. If τ is periodically identified, $\tau \sim \tau + \beta$, then the quantity $r_+ \tau$ ranges over an interval of length $r_+ \beta$. The (ρ, τ) -geometry is smooth at $\rho = 0$

if and only if $r_+\beta = 2\pi$, i.e. $\beta = 2\pi/r_+$. This is the standard Gibbons–Hawking argument: smoothness of the Euclidean cigar geometry fixes the temperature. The resulting geometry is the smooth Euclidean BTZ black hole, topologically a solid torus, with the thermal τ -circle contractible at the tip $\rho = 0$.

The microstate saddle. For the individual microstate $|P\rangle$, the situation is qualitatively different. The state is prepared by the path integral on the half-cylinder $D^2 \times (-\infty, 0]$ —the Euclidean time τ ranges over $(-\infty, 0]$ and is non-compact. There is no periodic identification of τ , and therefore the Gibbons–Hawking smoothness argument does not apply.

The (ρ, τ) -geometry near $\rho = 0$ is “polar coordinates” with an angular direction that ranges over $(-\infty, 0]$ rather than $[0, 2\pi)$. This is not a smooth point, not a cone, but a degenerate surface. In the Chern–Simons language, the origin of this degeneracy is clear: there is a Wilson line source at $\rho = 0$. The field strength has a delta-function singularity (essentially eq. (2.9), but with a hyperbolic holonomy source), localised at the centre of the disc, which is the horizon $r = r_+$ in the metric variables (4.11). In the Chern–Simons formulation this source is simply part of the classical equations of motion—the connection is smooth everywhere away from it. But in the metric language, the source shows up as a singularity at the horizon.

To summarise: in the microstate saddle, the Euclidean geometry covers the region $r \geq r_+$ (equivalently $\rho \geq 0$), with the ϕ -circle retaining a finite horizon perimeter $2\pi r_+$, but with no smooth contraction of the τ -direction. The “horizon” is the location of the Wilson line source—a singularity in the curvature, not a smooth interior point.

This is exactly the Euclidean geometry described in Section 4.2 of [2]: the centre of the disc carries a Wilson line in the Chern–Simons language, which translates to a finite “horizon area” (here, perimeter) in the metric language, with no smoothness condition because the Euclidean time direction need not be periodic.²⁸

4.4 Lorentzian Signature: Singular Horizon

We now Wick-rotate back to Lorentzian signature and examine the horizon of the individual microstate.

The Lorentzian metric (4.12) describes the *exterior* of the BTZ black hole, valid for $\rho > 0$ (i.e. $r > r_+$). Near the horizon $\rho \rightarrow 0^+$, the metric reduces to

$$ds^2 \approx -r_+^2 \rho^2 dt^2 + d\rho^2 + r_+^2 d\phi^2. \quad (4.15)$$

²⁸In the language of [2], the Wilson line plays the role of the inner boundary, and the presence of a source there is the gauge-theoretic avatar of the absence of a smooth horizon for an individual microstate.

The (ρ, t) -part is simply Rindler space – the near-horizon approximation familiar from any non-extremal black hole. Whether the Rindler horizon at $\rho = 0$ is a smooth, extendable surface or a genuine singularity depends on the *quantum state*.

The modular thermal state: smooth Kruskal extension. A *thermal state* by itself need not have a smooth horizon: the KMS condition can be imposed for any thermal ensemble. What is special about the black-hole state is the density of microstates. As we elaborate in Section 5, when the thermal sum is weighted by the density dictated by modular invariance, its dominant saddle satisfies $\beta r_+ = 2\pi$. The Euclidean thermal cycle is then smoothly contractible, and after analytic continuation the geometry extends through $\rho = 0$ to the full two-sided BTZ spacetime with a regular Kruskal horizon. Thus smoothness is a property of the modular thermal ensemble, rather than a consequence of KMS periodicity alone.

The individual microstate: singular horizon. The microstate $|P\rangle$ is a *pure state*, prepared by the Euclidean path integral on the non-compact half-cylinder. Its Euclidean preparation involves a Wilson line source at $\rho = 0$ —a delta-function in the gauge field curvature—and no periodic identification of Euclidean time. As a consequence:

1. The state does not satisfy KMS periodicity, nor does it have a modular density – it is a single eigenstate.
2. The Wilson line source at $\rho = 0$ is a genuine singularity that sources the CS field strength, not a coordinate artifact. Concretely, the CS field strength has a delta-function (source) at $\rho = 0$. This is a statement about the spatial slice—the punctured disc at any fixed time—and is therefore the same in both Euclidean and Lorentzian signatures.²⁹
3. The Lorentzian geometry of the microstate terminates at $\rho = 0$ (i.e. $r = r_+$) with a finite-perimeter. The horizon is not a smooth null surface through which the geometry can be extended³⁰.

²⁹The equation of motion (2.9) is a constraint on the spatial slice at any fixed time t (or τ). Since the spatial disc is the same regardless of whether the time direction is Lorentzian or Euclidean, the delta-function source at $\rho = 0$ carries over unchanged.

³⁰There is no contradiction with the fact that the exterior metric (4.12), considered by itself, admits the usual smooth BTZ Kruskal completion. That is a geometry designed to capture thermality. In the microstate saddle, $\rho = 0$ is the Wilson-line puncture, with a delta-function source for the Chern–Simons field strength and non-trivial holonomy around the linking ϕ -cycle. There is no natural notion of thermality, but there is a natural notion of a source sitting at $r = r_+$. The usual two-sided BTZ completion instead removes this source and changes the global completion of the spatial slice by extending through the would-be horizon to a second exterior. The same hyperbolic ϕ -holonomy is then supported by the non-contractible cycle of the source-free BTZ geometry. So we will view the existence of a smooth source-free completion as the answer to a different problem.

4.4.1 The Nature of the Singular Horizon

It is important to be precise about what kind of singularity the microstate horizon is. The Riemann tensor of the BTZ metric is that of locally AdS_3 everywhere—there is no curvature singularity in the usual Riemannian sense away from $\rho = 0$. The singularity is of a different character:

- In the **Chern-Simons formulation**, the connection is smooth and flat on the punctured disc $\rho > 0$, with a delta-function curvature source at $\rho = 0$. The gauge field description is completely regular away from the source.
- In the **metric formulation**, the delta-function in F_{ij} translates to a delta-function contribution to the curvature at $\rho = 0$ ³¹. The resulting singularity is analogous to the conical singularity at the tip of a conical defect (which also has distributional curvature supported at a point), but occurring now at a surface of finite perimeter.

An instructive comparison is with the sub-threshold (conical defect) case. For the conical defect microstate constructed in Section 2, the saddle metric is (2.5), with a conical singularity at $r = 0$ where the ϕ -circle collapses. The Wilson line source sits at the tip $r = 0$, producing a delta-function curvature that encodes the deficit angle. For the super-threshold (BTZ) microstate, the situation is qualitatively the same—a Wilson line source at the origin of the disc, producing a delta-function in the curvature—except that the ϕ -circle no longer collapses at the source³². Instead, it retains a finite circumference $2\pi r_+$, which is the horizon perimeter.

It is crucial to not forget that the singularity of the microstate is an artifact of the semi-classical (large- k) limit that we have taken on the quantum state. At any finite value of k , the state constructed by our Wilson line path integral is a perfectly well-defined Hilbert space state. But in the semi-classical limit, we find gauge fields with singular sources or metrics with singular horizons.

4.5 Smooth Horizon as an Ensemble of States

In both cases (black holes as well as microstates), the Chern-Simons holonomy around the spatial circle can be taken to be in the same hyperbolic conjugacy class, with the same boost parameter $\lambda = r_+$. The local geometry is the same—both are locally the BTZ metric.

³¹This translation assumes a compact thermal cycle. More generally, it can be viewed as being geodesically incomplete.

³²The word curvature can be read to mean either the CS field strength or the metric curvature in this sentence, but the collapse of the ϕ -circle is a metric reading.

In the BTZ black hole, Euclidean geometry is a solid torus $D^2 \times S^1_{2\pi}$ with periodic τ , with period β . (Note that the disc here is not the spatial disc.) There is no Wilson line: the disc is unpunctured. The saddle is the smooth Euclidean BTZ with the thermal τ -circle contractible at $\rho = 0$. Smoothness condition fixes $\beta = 2\pi/r_+$. In the Lorentzian geometry, this continues to the Hartle–Hawking state on the full two-sided BTZ spacetime. This is the Kruskal horizon. The holonomy is hyperbolic, boost parameter $\lambda = r_+$, fixed by the temperature (i.e. by the inverse temperature β).

We emphasise that the holonomy is determined by different physical inputs in the two cases. For the microstate, it is the state label s (equivalently, the conformal weight h_P) that fixes $\lambda = 2s/k$. For the thermal state, it is the temperature $T = 1/\beta$ that fixes λ via the smoothness condition $\beta = 2\pi/\lambda$. In the microstate, we specify the state, in the thermal case, we specify the temperature. The smooth horizon emerges only in the latter.

The situation when there is rotation, is similar. To specify the microstate, we now have to specify two holonomies, while to specify the ensemble we have to specify two potentials (the temperature and the angular velocity, or equivalently, left and right temperatures). The black hole can be understood as a grand-canonical partition function worth of such microstates. We will discuss this more in the next section. This picture provides a concrete, controlled realization of the idea that smooth horizons are emergent, coarse-grained features of thermal ensembles. Usually, we only have access to Gibbons-Hawking-like calculations which do not tell us what the individual microstates look like. Our point here is that one can construct these quantum microstates: considered individually, they have singular horizons in the semi-classical limit.

But we can do more: the thermal partition function is in fact a weighted sum over the microstates, with the (approximate) density of states dictated by modular invariance. Studying the thermal saddles and fluctuations of this partition function is instructive, and that is what we turn to, next.

5 Smoothness and Fluctuations

A thermal ensemble of microstates is useful as a statistical system if we have some understanding about *which* of these microstates are present in a given theory. In most situations that are of interest in statistical mechanics, particularly when strong chaos and thermalization are present, we do not know an exact answer to this question. But approximate knowledge can go a long way³³. In the case of holographic CFTs, in the high temperature limit, such an approximation becomes available to us thanks to modular invariance. In

³³Note for example, that we have a decent understanding of metals, despite the fact that they are almost impossibly complicated systems.

this regime, the vacuum in the dual channel is the dominant contribution to the partition function. Remarkably, the dual channel vacuum has a re-writing as a continuum sum over super-threshold (i.e., heavy) primaries in terms of a function called the modular S -kernel. This means that in the high temperature limit, the partition function is approximately a continuum sum over explicitly known microstates. This allows us to do and interpret some simple calculations and connect them to known physics. This is the goal of this section. The leading saddle will reproduce well-known results associated to the BTZ black hole, but we will also compute the fluctuations. This is a trivial computation for us, but its form will match the (somewhat complicated) “fluctuations of the area” type results in general relativity [57, 58].

A key fact about the above set up, is that it does *not* rely on the large- c limit. It is instead reliant on the Cardy regime for its validity, and therefore in principle, allows us to go beyond the leading large- c limit. In this paper our goals will be more modest: we want to connect with (semi-)classical gravitational expectations and therefore we will be working in the large- c (large- k) limit.

The perspective of this section is that we fix the temperature (and angular chemical potential), and the holonomy is determined dynamically as a saddle point of the ensemble integral. As we will show, the saddle automatically produces the smooth Euclidean BTZ—the smoothness of the thermal cycle is not imposed by hand but emerges as a saddle-point condition. Moreover, the Gaussian fluctuation around the saddle at finite k yields a variance $\text{Var}(S) = S$ for the Bekenstein–Hawking entropy, reproducing the BTZ version of “quantum fluctuations in the area” expected from perturbative general relativity.

5.1 From the Microstate to the Ensemble

Our discussion in the previous section was framed in the non-rotating BTZ setting, for simplicity of exposition. But the discussion is most naturally set up for a general rotating black hole, and since carrying the rotation along costs nothing, we do so throughout this section. The non-rotating statements are recovered by setting $\tau_1 = 0$ at the end.

The dictionary between the modular parameter $\tau = \tau_1 + i\tau_2$ of the boundary torus and the thermodynamic variables is,

$$\begin{aligned} q &= e^{2\pi i\tau}, & \bar{q} &= e^{-2\pi i\bar{\tau}}, \\ \beta &= 2\pi\tau_2, & \theta &= 2\pi\tau_1, \end{aligned} \tag{5.1}$$

with β the inverse temperature and θ the potential conjugate to the angular momentum $J = L_0 - \bar{L}_0$. Since the two chiral sectors will decouple, it is convenient to trade τ and $\bar{\tau}$ for

the left- and right-moving inverse temperatures

$$\beta_L = -2\pi i \tau = \beta - i\theta, \quad \beta_R = 2\pi i \bar{\tau} = \beta + i\theta, \quad (5.2)$$

so that $q = e^{-\beta_L}$, $\bar{q} = e^{-\beta_R}$ and $\beta = \frac{1}{2}(\beta_L + \beta_R)$. For a Lorentzian rotating black hole β_L and β_R are separately real and positive, which corresponds to imaginary τ_1 . We will not impose this, and treat τ and $\bar{\tau}$ as independent complex parameters throughout.

The grand-canonical partition function on the boundary torus is

$$Z(\tau, \bar{\tau}) = \text{Tr } q^{L_0 - c/24} \bar{q}^{\bar{L}_0 - c/24}, \quad (5.3)$$

and the object of interest for us is “the BTZ partition function”. This is the S -modular image of the vacuum character,

$$Z_{\text{BTZ}}(\tau, \bar{\tau}) = \chi_{\text{vac}}\left(-\frac{1}{\tau}\right) \bar{\chi}_{\text{vac}}\left(-\frac{1}{\bar{\tau}}\right), \quad (5.4)$$

which dominates Z at high temperature. This factorises into a holomorphic and an anti-holomorphic piece, so we present every computation in the holomorphic sector and obtain its counterpart by $\tau \rightarrow \bar{\tau}$, $P \rightarrow \bar{P}$, $\beta_L \rightarrow \beta_R$. When the black hole rotates the two sectors sit at different saddle points, and this is how the two horizon radii $r_{\pm}$ enter.

5.2 The S -Kernel as a Density of Primaries

The modular transform of the vacuum is an integral over non-degenerate Virasoro characters,

$$\chi_{\text{vac}}\left(-\frac{1}{\tau}\right) = \int_0^\infty dP \ S_{0P} \ \chi_P(\tau), \quad \chi_P(\tau) = \frac{q^{P^2}}{\eta(\tau)}, \quad (5.5)$$

where χ_P is the character of the Verma module built on the primary of weight $h_P = \frac{c-1}{24} + P^2$ of eq. (3.12), the factor $1/\eta(\tau)$ accounting for its Virasoro descendants. The kernel is [2, 13, 14]

$$S_{0P} = 4\sqrt{2} \sinh(2\pi bP) \sinh\left(\frac{2\pi P}{b}\right), \quad (5.6)$$

with b the Liouville parameter of eq. (3.12), $b^{-2} = k - 2$. In the gravity regime $k \gg 1$ we have $b \ll 1$, $Q \approx 1/b$ and $c \approx 6k$.

S_{0P} is the density of Virasoro primaries at momentum P dictated by modular invariance. Extracting the leading exponential from each $\sinh$ gives the exact rewriting

$$\log S_{0P} = 2\pi Q P + \log \sqrt{2} + \log \left[(1 - e^{-4\pi bP}) (1 - e^{-4\pi P/b}) \right], \quad (5.7)$$

which reduces to $\log S_{0P} \approx 2\pi QP$ once P is large compared to both b and $1/b$. Since $h_P \approx P^2$ for such heavy states and $Q^2 = \frac{c-1}{6}$, this is

$$\log S_{0P} \approx 2\pi \sqrt{\frac{c-1}{6}} h_P, \quad (5.8)$$

the Cardy growth of the primary degeneracy, with the shift $c \rightarrow c-1$ that separates primaries from descendants already built in. This is the sense in which the kernel supplies the microstates: it counts the heavy primaries of Section 3, each of which we have constructed as a Wilson-line state.

Assembling both chiralities,

$$Z_{\text{BTZ}}(\tau, \bar{\tau}) = \frac{1}{\eta(\tau) \bar{\eta}(\bar{\tau})} \int_0^\infty dP \int_0^\infty d\bar{P} S_{0P} S_{0\bar{P}} q^{P^2} \bar{q}^{\bar{P}^2}. \quad (5.9)$$

5.3 Saddle-Point Evaluation at Large k

All the P -dependence of (5.9) sits in the holomorphic spectral integral

$$I(\tau) \equiv \int_0^\infty dP S_{0P} e^{2\pi i \tau P^2} = \int_0^\infty dP S_{0P} e^{-\beta_L P^2}, \quad (5.10)$$

which we evaluate by steepest descent at large k . The exponent

$$f(P) = \log S_{0P} + 2\pi i \tau P^2 \quad (5.11)$$

is stationary at $f'(P_*) = 0$. Using

$$\frac{\partial}{\partial P} \log S_{0P} = 2\pi b \coth(2\pi b P) + \frac{2\pi}{b} \coth\left(\frac{2\pi P}{b}\right), \quad (5.12)$$

the saddle-point condition reads

$$2\pi b \coth(2\pi b P_*) + \frac{2\pi}{b} \coth\left(\frac{2\pi P_*}{b}\right) = -4\pi i \tau P_* = 2\beta_L P_*, \quad (5.13)$$

which is exact at any k .

The two arguments appearing here have a transparent meaning. With the exact relation $P = sb$ from eq. (3.11) and the leading semiclassical relation $\lambda = 2s/k + O(1/k)$ from eq. (3.15),

$$\frac{P_*}{b} = s_*, \quad 2b P_* = \frac{2s_*}{k-2} = \lambda_L (1 + O(1/k)), \quad (5.14)$$

so the second coth in (5.13) has argument $2\pi s_*$ and the first has argument $\pi \lambda_L (1 + O(1/k))$. For a macroscopic black hole $s_* \gg 1$, so the second coth is unity up to $O(e^{-4\pi s_*})$, while λ_L

is of order one, so the first term is $O(b)$ against the $O(1/b)$ of the second and is a relative correction of order $1/k$. This leads to,

$$\frac{2\pi}{b} = 2\beta_L P_* \quad \implies \quad P_* = \frac{\pi}{b\beta_L} = \frac{i}{2b\bar{\tau}}, \quad (5.15)$$

and in the antiholomorphic sector $\bar{P}_* = \pi/(b\beta_R) = -i/(2b\bar{\tau})$.³⁴

By (5.14), the boost parameters of the two spatial holonomies at the saddle are therefore, at leading order in $1/k$,

$$\lambda_L = 2bP_* = \frac{2\pi}{\beta_L} = \frac{i}{\bar{\tau}}, \quad \lambda_R = 2b\bar{P}_* = \frac{2\pi}{\beta_R} = -\frac{i}{\bar{\tau}}. \quad (5.16)$$

The horizon radii $r_{\pm} = \frac{1}{2}(\lambda_L \pm \lambda_R)$ of eq. (3.19) (we set AdS length $\ell = 1$ in all of this), are thus fixed by the temperature and the angular potential. In the non-rotating limit $\theta = 0$ we have $\beta_L = \beta_R = \beta$, $r_- = 0$ and $\lambda = r_+ = 2\pi/\beta$, while $s_* = P_*/b = \pi(k-2)/\beta$, which at leading order in large k is the identification $s = kr_+/2$ of eq. (3.15).

5.4 Smooth Horizon from the Saddle

The Euclidean geometry associated with the ensemble is a solid torus: the boundary torus of modular parameter τ , filled in so that the thermal cycle is contractible. The geometry is smooth where that cycle closes off precisely when the Chern-Simons holonomy around it is trivial (i.e., central in each of the two gauge groups). For the rotating BTZ solution this is the pair of conditions

$$\lambda_L \beta_L = 2\pi, \quad \lambda_R \beta_R = 2\pi, \quad (5.17)$$

equivalently $\beta_{L,R} = 2\pi/(r_+ \pm r_-)$. This is exactly what the saddle delivers in eq. (5.16), in both chiral sectors and for every τ .

In the metric language, the pair (5.17) is the pair of conditions that one usually imposes separately. Written in terms of β and θ , they are

$$\beta = \frac{2\pi r_+}{r_+^2 - r_-^2}, \quad i\theta = \beta\Omega_H, \quad \Omega_H = \frac{r_-}{r_+}, \quad (5.18)$$

or equivalently $\beta_{L,R} = \beta(1 \mp \Omega_H)$. The first is the absence of a conical defect at the outer horizon, which is the statement that $1/\beta$ is the Hawking temperature. The second fixes the twist of the thermal cycle: the cycle that closes off is the one generated by $\partial_t + \Omega_H \partial_\phi$, the Killing vector that becomes null on the horizon, and not by ∂_t . The factor of i is the remark

³⁴For general complex τ the saddle lies off the real axis, and the contour is deformed into the complex P -plane to pass through it. This is standard steepest descent, and the Gaussian analysis of Section 5.7 proceeds on the deformed contour exactly as it would on the real axis.

made below eq. (5.2), that τ_1 is imaginary for a Lorentzian rotating black hole, so both sides of the second relation are real. Only the first condition survives at $r_- = 0$, where $\theta = 0$ and $\beta = 2\pi/r_+$. In the Chern-Simons language, the two conditions turn into parallel statements about the triviality of the holonomy around the contractible cycle, imposed in each of the two gauge groups.

It is worth being explicit about what is and is not being assumed here. The microstate of Section 3 is specified through the pair of representation labels s_L and s_R , and its two spatial holonomies are $\lambda_{L,R} = 2s_{L,R}/k$ by eq. (3.19). In the semi-classical analysis of Section 4 the Euclidean time direction was non-compact, so that no smoothness condition was in effect: the geometry had a singular horizon. Here the temperature and the angular potential are specified instead, and the holonomies are obtained as saddles. In each chiral sector separately, the holonomy comes from the competition in (5.10) between the growth of the density of primaries, $S_{0P} \sim e^{2\pi QP}$, and the Boltzmann suppression $e^{-\beta_L P^2}$. The data is a pair of numbers in both cases, but the same two hyperbolic conjugacy classes are fixed by different physics:

$$\begin{aligned} \text{Microstate:} \quad \lambda_{L,R} &= \frac{2s_{L,R}}{k}, & \text{fixed by the representation labels,} \\ \text{Ensemble:} \quad \lambda_{L,R} &= \frac{2\pi}{\beta_{L,R}}, & \text{fixed by the saddle point.} \end{aligned} \quad (5.19)$$

The smoothness of the thermal cycle is a thermodynamic/statistical statement about the second, not a geometric input. This is the sense in which the smooth horizon is a property of the ensemble and not of the individual states that comprise it.

5.5 Lorentzian Smoothness, TFD and the Modular Ensemble

Because the answer is still the smooth horizon, it is easy to mistake that the saddle calculation is a trivial re-writing of the familiar Gibbons-Hawking calculation. So let us emphasize that it is quite different: it provides a statistical mechanics answer to black hole thermodynamics, while Gibbons-Hawking provided a geometric answer. The saddle balances the growth of the density of primaries against the Boltzmann factor, and the answer $\lambda_{L,R} = 2\pi/\beta_{L,R}$ came out of the specific exponential growth $S_{0P} \sim e^{2\pi QP}$ of the modular kernel. To clarify this, it is worth asking what would happen if that growth were different. Modular invariance fixes the kernel uniquely of course, but tuning it “by hand” is useful as a diagnostic: a way of identifying which feature of the spectrum the smoothness of the horizon is sensitive to.

Replace the kernel in (5.10) by a density of heavy primaries with a rescaled Cardy growth,

$$I_\gamma(\tau) = \int_0^\infty dP \, \rho_\gamma(P) e^{-\beta_L P^2}, \quad \rho_\gamma(P) = e^{2\pi\gamma QP}, \quad \gamma > 0, \quad (5.20)$$

so that $\gamma = 1$ is the modular value. The saddle-point condition is now $2\pi\gamma Q = 2\beta_L P_*$, so that $P_* = \pi\gamma Q/\beta_L$, and the boost parameter of the spatial holonomy at the saddle is $\lambda_L = 2bP_* = 2\pi\gamma/\beta_L$ at leading order in $1/k$, with the same statement in the antiholomorphic sector. In place of (5.17) one therefore finds

$$\lambda_L \beta_L = 2\pi\gamma, \quad \lambda_R \beta_R = 2\pi\gamma. \quad (5.21)$$

The holonomy around the contractible cycle is non-trivial unless $\gamma = 1$. In the metric language, by the near-horizon form (4.14), the (ρ, τ) plane now closes off with total angle $2\pi\gamma$ instead of 2π : the filled solid torus carries a conical defect at its core for $\gamma < 1$ and a conical excess for $\gamma > 1$, located at the horizon.

It is useful to say what γ measures. Since $h_P \approx P^2$ for heavy states and $Q^2 = \frac{c-1}{6}$, the growth in (5.20) can be written as in (5.8) with c replaced by an effective value,

$$\log \rho_\gamma \approx 2\pi \sqrt{\frac{c_{\text{eff}} - 1}{6}} h_P, \quad c_{\text{eff}} - 1 = \gamma^2 (c - 1). \quad (5.22)$$

A deformed density is thus one whose Cardy growth is governed by a central charge different from the c that fixes the Newton constant through $c \approx 6k$. This mismatch shows up in a second place as well. Evaluating (5.20) at its saddle by the steps of Section 5.6 gives $\log I_\gamma = \pi^2 \gamma^2 Q^2 / \beta_L$, so the entropy is γ^2 times its value at $\gamma = 1$, while the horizon radius $r_+ = \frac{1}{2}(\lambda_L + \lambda_R)$ is by (5.21) only γ times its value at $\gamma = 1$. At leading order in c the two do not agree,

$$S = \gamma \frac{A}{4G_N}, \quad (5.23)$$

which reduces to (5.30) at $\gamma = 1$. The Bekenstein-Hawking relation and the smoothness of the thermal cycle fail together, and both hold only at the modular value.

We can now state the Lorentzian version. For any density of states one may form the thermal density matrix at inverse temperature β and purify it in a doubled Hilbert space as the thermofield double,

$$|\text{TFD}\rangle \propto \sum_n e^{-\beta E_n/2} |n\rangle_L \otimes |n\rangle_R, \quad (5.24)$$

the sum running over the energy eigenstates of the boundary theory, so that the density of states enters through the multiplicities. This construction is kinematical: it is available for any spectrum and any temperature, and tracing out one factor returns the thermal state on the other. In the bulk, the Euclidean preparation of (5.24) is the path integral on half of the filled torus, cut along $\tau = 0$ and $\tau = \beta/2$, and the $t = 0$ slice of the resulting Lorentzian geometry is the Einstein-Rosen bridge joining the two boundaries [59, 60]. The bifurcation surface of that geometry is the core of the solid torus, the same locus that carries the conical defect above. It is a regular bifurcate Killing horizon, through which the geometry continues

to the full Kruskal extension, exactly when the Euclidean tip is smooth³⁵. By (5.21) this happens only at $\gamma = 1$.

Two statements that are often used interchangeably therefore come apart. The thermofield double is available for any spectrum. But not all TFDs correspond to an eternal Lorentzian black hole with a smooth horizon. When one writes the eternal black hole as a thermofield double and treats the horizon as a regular surface through which the geometry extends, one has implicitly assumed that the weights in (5.24) are the modular ones. We refer to the ensemble with that density of states as the *modular ensemble*. It is worth putting this beside the result of Section 4. There, an individual microstate was found to have a singular horizon, its holonomy being fixed by the representation label rather than by a saddle point. Here we see that averaging is not by itself the remedy: a thermal average taken with a non-modular density of primaries is singular at the horizon as well. Smoothness requires a regime where the modular ensemble is a good approximation to the spectrum. For BTZ, this is the Cardy regime.

5.6 On-Shell Action and the Bekenstein-Hawking Entropy

Before turning to fluctuations, we present the value of the integral at the saddle. By (5.14) the two hyperbolic cotangents in (5.13) have arguments $\pi\lambda_L(1+O(1/k))$ and $2\pi s_*$ there, and the last term of (5.7) involves the same two combinations. Both are large in the regime of interest, so up to exponentially small corrections the cotangents equal unity and that term is negligible. The saddle-point condition then gives $2\pi Q = 2\beta_L P_*$, so that $P_* = \pi Q/\beta_L$, which agrees with (5.15) up to a relative correction of order $1/k$, and $\log S_{0P_*} = 2\pi Q P_* + O(1)$. Using $2\pi i\tau = -\beta_L$,

$$\log I \approx f(P_*) = 2\pi Q P_* - \beta_L P_*^2 = \frac{2\pi^2 Q^2}{\beta_L} - \frac{\pi^2 Q^2}{\beta_L} = \frac{\pi^2 Q^2}{\beta_L} = \frac{\pi^2(c-1)}{6\beta_L}, \quad (5.25)$$

where the last step uses $c = 1 + 6Q^2$ from (2.30). This is the contribution of the primaries alone, and it is the Cardy free energy of a theory of central charge $c-1$, the integrated form of the primary counting (5.8).

The η -function prefactors in (5.9) carry the Virasoro descendants. With $\tau = i\beta_L/2\pi$, the modular transformation $\eta(-1/\tau) = \sqrt{-i\tau} \eta(\tau)$ gives

$$\eta(\tau) = \sqrt{\frac{2\pi}{\beta_L}} e^{-\pi^2/(6\beta_L)} \left(1 + O(e^{-4\pi^2/\beta_L})\right), \quad (5.26)$$

³⁵Another way to think about this is that Euclidean smoothness and Lorentzian smoothness are two faces of the same *analytic* condition here.

so that $\log(1/\eta(\tau)) = \pi^2/(6\beta_L)$ up to a term of order $\log\beta_L$. This is the Cardy free energy at $c = 1$, and adding the two pieces [2],

$$\log I + \log \frac{1}{\eta(\tau)} = \frac{\pi^2(c-1)}{6\beta_L} + \frac{\pi^2}{6\beta_L} = \frac{\pi^2 c}{6\beta_L}, \quad (5.27)$$

which realizes the split $(c-1) + 1 = c$ of Section 3 quantitatively. The antiholomorphic sector gives the same expression with $\beta_L \rightarrow \beta_R$, so

$$\log Z_{\text{BTZ}} \approx \frac{\pi^2 c}{6} \left(\frac{1}{\beta_L} + \frac{1}{\beta_R} \right), \quad (5.28)$$

the high-temperature free energy of a CFT of central charge c . The entropy of each chiral sector follows from (5.27) by the usual thermodynamic relation,

$$S_L = \left(1 - \beta_L \frac{\partial}{\partial \beta_L} \right) \left(\log I + \log \frac{1}{\eta} \right) = \frac{\pi^2 c}{3\beta_L} \approx 2\pi \sqrt{\frac{c h_*}{6}}, \quad h_* = P_*^2, \quad (5.29)$$

which is the Cardy formula [61], and similarly for S_R . Using $\lambda_{L,R} = 2\pi/\beta_{L,R}$ and $r_+ = \frac{1}{2}(\lambda_L + \lambda_R)$, the total entropy is

$$S = S_L + S_R = \frac{\pi c}{6}(\lambda_L + \lambda_R) = \frac{\pi c r_+}{3} = \frac{2\pi r_+}{4G_N}, \quad (5.30)$$

the Bekenstein-Hawking entropy of the rotating BTZ black hole, with $2\pi r_+$ the horizon perimeter. In the fluctuation analysis of the next two subsections we work to leading order in c , where the distinction between c and $c-1$ plays no role.

5.7 Gaussian Fluctuations Around the Saddle

The saddle point of Section 5.3 gives the leading thermodynamics. Because we have an explicit statistical description, we can go further and compute the fluctuations of the ensemble. Writing $P = P_* + \delta P$ and expanding (5.11),

$$f(P) = f(P_*) + \frac{1}{2} f''(P_*) (\delta P)^2 + \mathcal{O}((\delta P)^3), \quad f''(P_*) = \frac{\partial^2}{\partial P^2} \log S_{0P} \Big|_{P_*} - 2\beta_L, \quad (5.31)$$

where we used $4\pi i\tau = -2\beta_L$. Differentiating (5.12) once more,

$$\frac{\partial^2}{\partial P^2} \log S_{0P} = -\frac{(2\pi b)^2}{\sinh^2(2\pi b P)} - \frac{(2\pi/b)^2}{\sinh^2(2\pi P/b)}. \quad (5.32)$$

The arguments at the saddle are $\pi\lambda_L(1 + O(1/k))$ and $2\pi s_*$, as in (5.14). The second term is therefore exponentially small in s_* , and the first is $O(b^2) = O(1/k)$, so both are negligible against $2\beta_L$ and

$$-f''(P_*) = 2\beta_L (1 + O(1/k)). \quad (5.33)$$

The Gaussian integral around the saddle is then

$$I \approx e^{f(P_*)} \sqrt{\frac{2\pi}{-f''(P_*)}} = e^{f(P_*)} \sqrt{\frac{\pi}{\beta_L}}, \quad (5.34)$$

and the spread of the Liouville momentum about the saddle is

$$\langle (\delta P)^2 \rangle = \frac{1}{-f''(P_*)} = \frac{1}{2\beta_L}. \quad (5.35)$$

Since $h_P = \frac{c-1}{24} + P^2$ by (3.12), the additive constant drops out of the variance and the fluctuation of the conformal weight is the fluctuation of P^2 . With $P^2 - P_*^2 = 2P_* \delta P + (\delta P)^2$, the first term dominates the second by a factor of order $\sqrt{k}$, so

$$\text{Var}(h) = 4P_*^2 \langle (\delta P)^2 \rangle = \frac{2P_*^2}{\beta_L} = \frac{\pi^2 c}{3\beta_L^3}, \quad (5.36)$$

using $P_* = \pi Q/\beta_L$ from (5.25) and $c-1 = 6Q^2$, so that $P_*^2 = \pi^2(c-1)/(6\beta_L^2)$, which is $\pi^2 c/(6\beta_L^2)$ at leading order in $1/k$. The antiholomorphic sector is independent, and the same steps give $2\bar{P}_*^2/\beta_R = \pi^2 c/(3\beta_R^3)$ for the corresponding variance there.

It is worth noting that (5.36) also follows without expanding around the saddle. The spectral integral (5.10) is a Boltzmann sum with β_L conjugate to P^2 , so the cumulants of P^2 are derivatives of $\log I$ with respect to β_L ,

$$\langle P^2 \rangle = -\frac{\partial}{\partial \beta_L} \log I, \quad \text{Var}(h) = \frac{\partial^2}{\partial \beta_L^2} \log I, \quad (5.37)$$

and inserting $\log I \approx \pi^2(c-1)/(6\beta_L)$ from (5.25) gives $\text{Var}(h) = \pi^2(c-1)/(3\beta_L^3)$, which is (5.36) at leading order in $1/k$. Had we instead used $\log(I/\eta)$, the same steps would return $\text{Var}(L_0)$, with the descendants restoring $c-1 \rightarrow c$.

5.8 Fluctuations of the Entropy and the Horizon Area

To turn (5.36) into a statement about the entropy we need the response of S_L to a change in h . From the Cardy form $S_L = 2\pi\sqrt{ch}/6$ of (5.29),

$$\left. \frac{\partial S_L}{\partial h} \right|_{h_*} = \frac{S_L}{2h_*} = \beta_L, \quad (5.38)$$

which is the first law for the chiral sector. The fluctuations are Gaussian and small, so linear response is enough, and

$$\text{Var}(S_L) = \left(\frac{\partial S_L}{\partial h} \right)^2 \text{Var}(h) = \beta_L^2 \frac{\pi^2 c}{3\beta_L^3} = \frac{\pi^2 c}{3\beta_L} = S_L, \quad (5.39)$$

the last step by (5.29). The right-moving sector gives $\text{Var}(S_R) = S_R$ in the same way. The two chiral integrals in (5.9) factorise, so the sectors fluctuate independently and their variances add,

$$\text{Var}(S) = \text{Var}(S_L) + \text{Var}(S_R) = S_L + S_R = S. \quad (5.40)$$

The variance of the Bekenstein-Hawking entropy is the entropy itself. This is a Poisson-like relation, and it holds for the rotating black hole as it stands, with S the total entropy (5.30).

The relation is not an accident of the S -kernel. Each chiral integral is a Boltzmann sum with β_L conjugate to h , so $\text{Var}(S_L) = \beta_L^2 \text{Var}(h)$ is the heat capacity of that sector, and a free energy of the Cardy form $\log I \propto 1/\beta_L$ has $C_L = S_L$. The two sectors combine into the familiar thermodynamic statement: with $E = h_L + h_R$ and $J = h_L - h_R$ as in (3.18), and using $\beta_{L,R} = \beta(1 \mp \Omega_H)$ from (5.18),

$$\delta S = \beta_L \delta h_L + \beta_R \delta h_R = \beta (\delta E - \Omega_H \delta J), \quad (5.41)$$

which is the first law of the rotating black hole, and (5.40) is the statement that the heat capacity at fixed angular velocity equals the entropy. What the computation above supplies is the microscopic content: the fluctuating variable is the Liouville momentum, which is the label of the Wilson-line microstate and therefore of the holonomy.

Because the entropy is the horizon perimeter in Planck units, this translates immediately into a statement about the area. With $S = A/4G_N$ and $A = 2\pi r_+$,

$$\text{Var}(A) = (4G_N)^2 \text{Var}(S) = 4G_N A, \quad \frac{\Delta A}{A} = \sqrt{\frac{4G_N}{A}} = \frac{1}{\sqrt{S}}. \quad (5.42)$$

The same answer can be read off directly from the holonomy, which makes the origin of the fluctuation explicit. At the leading order in $1/k$ used here, by (5.14) the horizon radius $r_+ = \frac{1}{2}(\lambda_L + \lambda_R) = b(P + \bar{P})$ is a linear function of the two Liouville momenta, so (5.35) gives

$$\text{Var}(A) = 4\pi^2 b^2 \left[\langle (\delta P)^2 \rangle + \langle (\delta \bar{P})^2 \rangle \right] = 2\pi^2 b^2 \left(\frac{1}{\beta_L} + \frac{1}{\beta_R} \right), \quad (5.43)$$

and since $A = 2\pi^2(\beta_L^{-1} + \beta_R^{-1})$ by (5.16), this is $4G_N A$ once $b^2 \approx 1/k$ and $4G_N = 1/k$ are used. The area fluctuation is the spread of the boost parameter of the spatial holonomy in the ensemble.

At the same leading order, the inner horizon is $r_- = \frac{1}{2}(\lambda_L - \lambda_R) = b(P - \bar{P})$, so the same two variances give the full covariance matrix of the radii,

$$\text{Var}(r_{\pm}) = \frac{b^2}{2} \left(\frac{1}{\beta_L} + \frac{1}{\beta_R} \right), \quad \text{Cov}(r_+, r_-) = \frac{b^2}{2} \left(\frac{1}{\beta_L} - \frac{1}{\beta_R} \right). \quad (5.44)$$

The covariance vanishes at $\theta = 0$. The non-rotating ensemble has $\langle J \rangle = 0$ while J still fluctuates, and it does so independently of the entropy. To Gaussian order, $\text{Var}(S) = S$ is

then unchanged if one further restricts to $J = 0$, so the non-rotating results of this section are limits of the rotating ones at the level of the fluctuations and not only at the level of the saddle.

The fractional fluctuation $\Delta A/A = 1/\sqrt{S}$ is small for a macroscopic black hole and becomes of order one when S does. Since $S = 2\pi k r_+$, the relative variance is $\text{Var}(S)/S^2 = 1/S \sim 1/(k r_+)$, so for r_+ of order one the fluctuations are a one-loop effect in the Chern-Simons expansion, as one would expect of a quantum correction³⁶.

Comparison with semi-classical gravity. Relations of the form $\text{Var}(A) \propto G_N A$ have been obtained recently by quite different routes. Parikh and Pereira [57] computed the renormalised graviton propagator in linearised quantum gravity on the four-dimensional Schwarzschild background in the Hartle-Hawking state, and found $\Delta A \sim r_H l_P$, with r_H the Schwarzschild radius and $l_P = \sqrt{G\hbar}$ the Planck length, which is $\text{Var}(A) \sim G A$. Ciambelli, He and Zurek [58] obtained a lower bound of the same parametric form for a finite causal diamond in $(d + 2)$ -dimensional Minkowski spacetime, from the covariant phase space of a stretched horizon.

These are not calculations of the same observable, and the $O(1)$ coefficients are not directly comparable. But they are suggestive. The graviton computation holds the coordinate sphere $r = r_H$ fixed and lets its shape fluctuate, so that the horizon radius does not itself fluctuate, while in the ensemble above the area varies because the boost parameter of the spatial holonomy, and with it r_+ , varies across the microstates. Ours is moreover a variance at fixed β_L and β_R , that is, at fixed temperature and angular potential. What the present setting supplies is a statistical-mechanical reading of the area fluctuation. The fluctuating variable is the label of an explicit Wilson-line microstate, and the variance is a spread over the same ensemble whose counting gives the entropy. Parikh and Pereira compute the variance of the area operator in a single fixed state, and observe that its relation to fluctuations of the black hole entropy remains unclear, since in the absence of a non-perturbative holographic description the area does not carry a statistical-mechanical interpretation as entropy [57]. Note also that 4D Schwarzschild, with its negative specific heat, is a transient object and is therefore not the easiest place to formulate these statements.

³⁶There is no tension with the fluctuations being “thermal”: because $S \sim k$ plays the role of $1/\hbar$, the thermal spread of the ensemble is automatically of one-loop size. Equivalently, in the purification (5.24) of Section 5.5, the same variance is the quantum-mechanical variance of a one-sided observable in the pure state $|\text{TFD}\rangle$.

5.9 Thermal AdS_3 : A Wilson Tower of Conical Defects

Our discussion so far naturally involved only states above the BTZ threshold, which is loosely the same as being in the Cardy regime: we placed the microstates in the (grand-) canonical ensemble and noticed that the result has a geometric interpretation in terms of smoothness of the saddle configuration. It is natural to wonder if there are any statements one can make regarding microstates below the BTZ threshold. This is not the regime that is under the jurisdiction of Cardy, so one might think that there is nothing much we can say about it. But while the kinematics of modular invariance does not seem directly instructive here, dynamical statements about the actual spectrum of the CFT can teach us a few things.

The key dynamical statement is that the spectrum of a holographic CFT seems often to be organized in terms of local (generalized free) fields in the bulk, rather than in terms of a structureless spectrum of isolated conical defects. In large- N gauge theories, this is of course naturally realized in terms of the single- and multi-trace sectors. It was recently noted in [6], building on the works of [62, 63, 54], that a bulk local field is naturally viewed as a tower of conical defect Wilson lines organized in terms of a seed primary and its associated multi-trace composites. This picture provides a bulk interpretation for the one loop determinant of the thermal gas of “gravitons” (more generally, light fields) one expects below the Hawking-Page transition: it is simply the sum of thermal traces of each primary in the multi-trace tower of Wilson lines [6].

Acknowledgments

We thank Apoorva Asthana, ChatGPT/Claude, Vishal Gayari and Pradipta Pathak for discussions. This work is a natural follow-up of the ideas in [2]. Some (though not all) of the conclusions here have already been reported earlier, scattered across the discussions in [2, 6, 36]. A key technical ingredient here is that we have made things more explicit in terms of quantum Drinfel’d-Sokolov reduction. We have also provided the details of some of the derivations, which were previously simply mentioned as results.

A States from Path Integrals: The $SU(2)_k$ Prototype

This appendix gives a self-contained, pedagogical account of the path-integral state construction used in the main text, but in the compact setting of $SU(2)_k$ Chern-Simons theory. This is in principle entirely well-known, but we try to emphasize points that we could not easily decipher from the literature. Two questions are addressed. First, what does a Euclidean path integral on a half-infinite geometry prepare when the bulk Hamiltonian vanishes, as it

does in a topological theory? The familiar slogan – “Euclidean evolution projects onto the lowest-energy state” – is a statement about gapped systems, and it needs to be replaced suitably. Second, what exactly does the semi-infinite Wilson line prepare on the punctured disc? We will see that the invariant answer is a *multiplet*-valued affine primary: the affine-primary condition $J_{n>0}^a \Psi = 0$ is a theorem, while the selection of a particular vector (e.g. the highest-weight state) is a choice of boundary data at the far endpoint of the Wilson line. This sharpened statement is what transfers to the noncompact gravitational problem, and it is what makes the continuous-series construction of Section 3 – where no extremal vector exists – structurally identical to the discrete-series one.

The gravitational $SL(2, \mathbb{R})$ problem of the main text differs from this prototype in some ways, especially because the physical Brown-Henneaux Virasoro algebra is obtained only after Drinfel’d-Sokolov reduction. The compact example should therefore be read as a prototype for the punctured-disc Hilbert space and for the state-preparation logic, not as a literal model of the full gravitational reduction.

A.1 Half-line Path Integrals in Quantum Mechanics

Consider first ordinary quantum mechanics with Hamiltonian H . The Euclidean path integral on $\tau \in [-T, 0]$, with the boundary value $q(0) = q$ fixed and with initial data described by a state $|\chi\rangle$ at $\tau = -T$, computes

$$\Psi_T(q) = \langle q | e^{-TH} | \chi \rangle. \quad (\text{A.1})$$

If H has a unique ground state ψ_0 separated by a gap, then as $T \rightarrow \infty$

$$\Psi_T(q) \longrightarrow e^{-TE_0} \psi_0(q) \langle \psi_0 | \chi \rangle, \quad (\text{A.2})$$

so that, up to normalization, the prepared state is ψ_0 regardless of $|\chi\rangle$. This is the textbook statement that the half-line path integral “prepares the ground state.” It is worth being precise about what this means: the far-end data $|\chi\rangle$ never disappears from the formula. It is merely damped into an overall constant. Two pieces of small print matter for us:

1. If the minimal-energy subspace is *degenerate*, the projection lands in that subspace, but the direction *within* it remains entirely determined by $|\chi\rangle$ – i.e., by the boundary data at the far end.
2. If $H = 0$, then $e^{-TH} = \mathbf{1}$ and (A.1) is just the overlap $\langle q | \chi \rangle$. Nothing is projected: the state prepared *is* the far-end data, transported to the cut. This is the case in topological theories of interest to us.

One might hope to evade the choice of $|\chi\rangle$ by “specifying nothing” at $\tau = -T$. But operationally, specifying nothing can only mean integrating over the far endpoint $q(-T) = q'$ with some measure $\mu(q')$, and

$$\int dq' \mu(q') \langle q | e^{-TH} | q' \rangle = \begin{cases} \psi_0(q) \int dq' \mu(q') \psi_0^*(q') \cdot e^{-TE_0} & (\text{gapped, } T \rightarrow \infty), \\ \mu(q) & (H = 0). \end{cases} \quad (\text{A.3})$$

In the gapped case the μ -dependence collapses into normalization. In the $H = 0$ case the wavefunction one prepares *is* the measure one chose at infinity. The same conclusion holds for first-order (phase-space) systems: for $S = \int d\tau (-i p \dot{q} + H)$ with $H = 0$, the kernel over any Euclidean interval is $\langle q | q' \rangle = \delta(q - q')$, and the half-line p - q path integral returns the input wavefunction unchanged.

The doctrine that survives all cases is this: *a path integral on a manifold with boundary defines a wavefunctional of the boundary data, and the state it represents is fixed by (i) the insertions and (ii) the conditions supplied at the other ends of the manifold.*

A.2 The Punctured-disc Hilbert Space

Now consider $SU(2)_k$ Chern-Simons theory on $D^2 \times \mathbb{R}_\tau$, with a Wilson line in the spin- j representation V_j inserted at the origin of the disc and running along the Euclidean time direction. At fixed τ , the spatial slice is a disc with one puncture, and the puncture carries the label j .

The CS/WZW correspondence states that the Hilbert space of Chern-Simons theory on a surface with boundary is described by the corresponding chiral WZW conformal blocks on that surface [64, 65]. In the present case, the boundary of the disc is a circle carrying the chiral $SU(2)_k$ WZW degrees of freedom, and the puncture selects the affine sector. Thus

$$\mathcal{H}(D^2; j) \cong \widehat{V}_j, \quad (\text{A.4})$$

where $\widehat{V}_j$ is the integrable highest-weight module of $\widehat{\mathfrak{su}}(2)_k$ built on the finite-dimensional spin- j representation V_j .

In radial quantization of the boundary WZW model, this statement is the state-operator correspondence [66]. The WZW primary Φ_j associated with the puncture is a V_j -valued field: its components $\Phi_{j,m}$, $m = -j, \dots, j$, inserted at the origin create the degenerate multiplet of affine primary states $|j, m\rangle$, and the negative current modes generate the rest of the affine module:

$$J_{-n_1}^{a_1} \cdots J_{-n_p}^{a_p} |j, m\rangle, \quad n_i > 0.$$

For compact $SU(2)_k$, integrability restricts the allowed labels to

$$j \in \left\{ 0, \frac{1}{2}, 1, \dots, \frac{k}{2} \right\}. \quad (\text{A.5})$$

Thus there is a finite set of allowed puncture labels, and each label selects a corresponding affine module. Note that (A.4) is an isomorphism of Hilbert spaces, i.e. a statement about *which* states exist on the punctured disc. It does not by itself say which vector a given path integral prepares. That is the question we turn to next.

A.3 The Half-cylinder State: Endpoint Data and the Primary

Perform the Euclidean path integral on the half-cylinder

$$M_- = D^2 \times (-\infty, 0], \quad (\text{A.6})$$

with the Wilson line running from $\tau = -\infty$ to $\tau = 0$ along the core $\{0\} \times (-\infty, 0]$. As mentioned already, an *open* Wilson line is not a number: it is the path-ordered exponential $\mathcal{P} \exp \int A$, a V_j -endomorphism-valued functional of the connection, with a free representation index at each end. The index at $\tau = 0$ is part of the definition of the states on the punctured disc. The index at $\tau = -\infty$, however, is free: under a gauge transformation it rotates in V_j , so a well-defined (gauge-covariantly meaningful) state is obtained only after it is contracted with a vector $v \in V_j$. The half-cylinder path integral is therefore, naturally viewed as a *linear map*

$$\Psi_j : V_j \longrightarrow \mathcal{H}(D^2; j), \quad v \longmapsto |\Psi_j(v)\rangle = \int [\mathcal{D}A] W_j(\{0\} \times (-\infty, 0]; v) e^{-S_{\text{CS}}^E[A]} |A|_{\tau=0}\rangle, \quad (\text{A.7})$$

where $W_j(\gamma; v)$ denotes the open Wilson line with its far endpoint contracted with v ³⁷. This is the Chern-Simons incarnation of the $H = 0$ lesson of Appendix A.1.

Although the bulk theory is topological – the bulk Hamiltonian is a constraint, so there is no bulk analogue of the damping in (A.2) – the *edge* theory is not. Once the boundary polarization is fixed, the boundary circle carries the chiral WZW model, and Euclidean translation along the semi-infinite boundary cylinder is generated by the edge Sugawara operator L_0^{Sug} , whose eigenvalue on the affine primaries of the spin- j sector is

$$h_j^{SU(2)} = \frac{j(j+1)}{k+2}, \quad (\text{A.8})$$

³⁷One might again try to dodge the choice of v by averaging suitably. But averaging the endpoint over the group with the invariant (Haar) measure projects v onto the invariant subspace $\text{Inv}(V_j)$, which vanishes for every $j \neq 0$. For a nontrivial representation, “democratically summing over the far-end data” does not produce a natural state – it produces zero. A charged endpoint cannot be gauge-invariantly hidden.

and $h_j^{SU(2)} + n$ on level- n current descendants. The factor $e^{-TL_0^{\text{Sug}}}$, $T \rightarrow \infty$, therefore genuinely damps: it removes every current descendant relative to the primaries. But it can go no further. L_0^{Sug} commutes with the zero modes J_0^a , so all $2j+1$ affine primaries $|j, m\rangle$ are *exactly degenerate* at (A.8), and the Euclidean damping is blind to the m -direction. We are squarely in case 1 of Appendix A.1: the projection lands on the degenerate primary multiplet, and the vector within it is fixed by the far-end data – here, the endpoint contraction v .

Putting the two observations together yields a uniqueness statement: the map (A.7) is completely fixed by symmetry, up to a single overall constant. The argument has three steps.

Step 1: the image lies in the affine-primary subspace of $\widehat{V}_j$. This is the content of the damping argument just given: whatever v is fed in, the output state has components only along the span of the $2j+1$ primaries $|j, m\rangle$ annihilated by all $J_{n>0}^a$. (The same statement will be re-derived, using boundary CFT methods and with no appeal to energetics, in Appendix A.4.)

Step 2: the map intertwines the zero-mode $\mathfrak{su}(2)$ action. By this we mean that rotating the input vector and then performing the path integral gives the same answer as performing the path integral and then rotating the output state with the boundary charges. To see this, consider a rigid (constant) gauge transformation $g \in SU(2)$ on the half-cylinder. Constant transformations are compatible with the boundary polarization: they survive as the global symmetry of the edge theory, implemented on the edge Hilbert space by the charges J_0^a . The Chern-Simons action and the measure are invariant, so the path integral is unchanged if one simultaneously transforms everything that carries a gauge index. There are exactly two such objects: the free Wilson-line endpoint at $\tau = -\infty$, where $v \rightarrow g \cdot v$ in V_j , and the boundary data at $\tau = 0$, on which g acts through the global charge. Invariance under doing *both* equates the two actions on the output:

$$|\Psi_j(g \cdot v)\rangle = U(g) |\Psi_j(v)\rangle, \quad U(g) = e^{i\theta_a J_0^a}. \quad (\text{A.9})$$

This is nothing but the Ward identity for global gauge rotations. (Looking ahead slightly: a local avatar of this appears in Appendix A.4. The $n=0$ contour integral picks up the simple pole at the endpoint insertion and acts as the generator t^a on the dangling index – the zero modes measure the endpoint charge.)

Step 3: Schur's lemma. On the affine-primary subspace the zero modes J_0^a act exactly as the spin- j matrices t^a act on V_j : the grade-zero floor of the module is the horizontal representation. So, by Steps 1 and 2, the path integral is a linear map between two irreducible copies of V_j that commutes with the $\mathfrak{su}(2)$ action, and Schur's lemma forces it to be a multiple of the identity.³⁸

³⁸Step 1, where the restriction to the primary subspace happens, is the important step. The full module

The map is therefore canonical: it comes with no choices other than which vector v is fed into it. This gives the invariant, choice-free statement of what the path integral constructs:

The undecorated semi-infinite Wilson-line path integral is the canonical embedding $V_j \hookrightarrow \widehat{V}_j$ of the zero-mode multiplet onto the affine-primary subspace, unique up to normalization. A state is obtained only after contracting the free endpoint index: the contraction $v = |j, m\rangle$ prepares the affine primary $|j, m\rangle$.

In particular, choosing the highest-weight vector $v = |j, \text{hw}\rangle$ gives

$$|\Psi_j\rangle = |j; \text{hw}\rangle \in \widehat{V}_j, \quad (\text{A.10})$$

which satisfies

$$J_n^a |j; \text{hw}\rangle = 0 \quad \forall n > 0, \quad J_0^+ |j; \text{hw}\rangle = 0, \quad J_0^3 |j; \text{hw}\rangle = j |j; \text{hw}\rangle. \quad (\text{A.11})$$

The logical status of these conditions is not uniform, and keeping them apart is the main point of this appendix. The first condition – the affine-primary condition – is automatic: it holds for every choice of v . The remaining two express the *choice* $v = |j, \text{hw}\rangle$. For $SU(2)$ this is a natural choice, since the highest-weight vector is distinguished within the unitary multiplet. But it is a prescription supplied at $\tau = -\infty$, not something that came from dynamics. Any presentation of the half-cylinder path integral that yields a specific vector without mentioning a choice has smuggled the contraction in somewhere, e.g. through a worldline gauge-fixing at $\tau \rightarrow -\infty$ or through an $i\epsilon$ -type regulator that damps onto the extremal weight. These are legitimate prescriptions, they are not outputs.

A.4 Regularity at the Puncture: Contour Argument

The affine-primary condition can be derived in a way that makes its kinematic character manifest. Map the semi-infinite boundary cylinder to the punctured unit disc by

$$z = e^{\tau + i\phi}, \quad (\text{A.12})$$

so that the $\tau = 0$ boundary circle becomes $|z| = 1$ and the far end $\tau = -\infty$ becomes the origin $z = 0$. In this frame the entire representation content of the Wilson line is concentrated at $z = 0$ as the V_j -valued primary insertion $\Phi_j(0)$ of Appendix A.2, and the state on $|z| = 1$

$\widehat{V}_j$ contains many further spin- j copies at higher grades – the current descendants also organize into $\mathfrak{su}(2)$ multiplets – so an intertwiner $V_j \rightarrow \widehat{V}_j$ is *not* unique by itself. Uniqueness holds because the geometry independently forces the image onto grade zero.

is the disc one-point conformal block with that insertion. Acting with a current mode on the state means inserting the contour integral

$$J_n^a = \oint \frac{dz}{2\pi i} z^n J^a(z) \quad (\text{A.13})$$

on a circle just inside the boundary. Away from insertions the current is holomorphic – this is the chiral Ward identity, and in bulk language it is Gauss’s law: flatness ($F = 0$) allows the circle on which the charge is measured to be deformed freely. The contour can therefore be slid inward until it hits the puncture at $z = 0$.

What does the puncture look like to the current? A *bare* Wilson-line endpoint is a pointlike source of charge in the representation V_j . The most singular thing the current can do near it is measure that charge:

$$J^a(z) \Phi_j(0) \sim \frac{(t^a \Phi_j)(0)}{z} + \text{regular}, \quad (\text{A.14})$$

a simple pole whose residue is the spin- j generator t^a acting on the dangling endpoint index. Higher-order poles would signal additional structure attached to the endpoint – derivative dressings, i.e. current descendants stuck to the tip. “Undressed endpoint $\Rightarrow$ at most a simple pole” is the precise content of “the state is an affine primary,” and it is a statement about what was inserted. Evaluating the moments,

$$J_n^a \Psi \longleftrightarrow \oint \frac{dz}{2\pi i} \left[z^{n-1} (t^a \Phi_j)(0) + O(z^n) \right], \quad (\text{A.15})$$

one reads off:

- $n \geq 1$: the integrand is holomorphic at the origin (z^{n-1} with $n-1 \geq 0$), so there is no residue and $J_{n \geq 1}^a \Psi = 0$;
- $n = 0$: the residue is $(t^a \Phi_j)(0)$ – the zero modes act as the ordinary finite-dimensional generators on the endpoint index, rotating the multiplet;
- $n \leq -1$: the integrand has genuine higher-order poles; these modes do not vanish, and create the current descendants.

A loose physical picture is: the modes J_n^a are the multipole moments of the edge charge distribution, weighted by z^n . The Wilson line presents the edge with a point charge sitting exactly at the origin, and every positive moment of a point charge at the origin vanishes. The zeroth moment is the total charge, which is exactly the $\mathfrak{su}(2)$ action on the multiplet.

A.5 Affine Tower and Lessons for Gravity

The full affine module contains the current-algebra descendants

$$J_{-n_1}^{a_1} J_{-n_2}^{a_2} \cdots J_{-n_p}^{a_p} |j, m\rangle, \quad n_i > 0, \quad (\text{A.16})$$

with Sugawara weight $h_j^{SU(2)} + \sum_i n_i$. These are the states preparable by *decorated* half-cylinders – inserting boundary current modes at $\tau < 0$. The module $\widehat{V}_j$ of (A.4) is the answer to the question “what states exist on the punctured disc”. The reachable set of the *undecorated* path integral (A.7) on the other hand, is only the degenerate horizontal multiplet $V_j \subset \widehat{V}_j$. All of these states share the same puncture label j . Using this compact example as an analogy for the gravitational problem, this fixed label is analogous to fixing the primary holonomy data.

We conclude with some comments on the gravitational case. First, Sugawara gives a Virasoro action on the affine module, but it does not identify the affine module one-to-one with the Virasoro descendant module of a single primary. The affine module is generally larger. In the gravitational $SL(2, \mathbb{R})$ problem one must in addition impose the Drinfel’d-Sokolov reduction associated with the Brown-Henneaux boundary conditions, and only after that reduction should the surviving physical sector be interpreted as a Virasoro primary together with its Virasoro descendants.

Second – the half-cylinder canonically prepares the *affine-primary multiplet* (as the intertwiner $V_j \hookrightarrow \widehat{V}_j$), and a particular vector only after an endpoint contraction is chosen. For compact $SU(2)$ the unitary multiplet has a distinguished extremal vector (the highest-weight state, annihilated by J_0^+ with J_0^3 -eigenvalue $+j$). In the noncompact discrete series $\mathcal{D}_j^+$ of the main text, there is again a distinguished extremal vector – now the lowest-weight state, annihilated by J_0^- with the same eigenvalue $+j$. In the principal continuous series of Section 3, there is no extremal vector at all. But DS reduction ensures that none is needed. In every case the invariant content of the path integral is the multiplet-valued affine primary. In the gravitational problem the endpoint data are unreduced affine data, but they are redundant because of the reduction. The Whittaker/zero-mode analysis of Section 2.5.3 and Appendix B shows that, at fixed Casimir, the BRST complex has a single level-zero class in the discrete and continuous cases alike.

B Principal-Series Zero Mode and BRST Cohomology

This appendix presents the only representation-dependent ingredient of the principal-series reduction that is needed in the main text: the level-zero Brown-Henneaux constraint selects a single primary class at fixed Casimir. The full positive-level BRST cohomology is deferred to [33].

The unreduced horizontal representation is the principal continuous series $\mathcal{C}_s^\epsilon$, with

$$J_0^3 |s, \epsilon; m\rangle = m |s, \epsilon; m\rangle, \quad m \in \epsilon + \mathbb{Z}, \quad C_2 = s^2 + \frac{1}{4}. \quad (\text{B.1})$$

It has no extremal vector. In a standard weight basis one may write

$$J_0^- |s, \epsilon; m\rangle = B_m |s, \epsilon; m-1\rangle, \quad (\text{B.2})$$

where $B_m \neq 0$ for every allowed m when $s > 0$. The precise normalization of B_m will not be needed.

As explained in Section 2.5.1, Brown-Henneaux boundary conditions impose a nonzero constant on the boundary cylinder. Keeping its normalization temporarily as $t \neq 0$, the level-zero part of the flowed BRST constraint is

$$(J_0^- - t)|\psi\rangle = 0. \quad (\text{B.3})$$

The reduction is naturally formulated in the formal direct-product completion which can be written as $\prod_{m \in \epsilon + \mathbb{Z}} \mathbb{C} |s, \epsilon; m\rangle$ of the zero-mode direction (with no growth condition)³⁹, so we can write

$$|\psi\rangle = \sum_{m \in \epsilon + \mathbb{Z}} c_m |s, \epsilon; m\rangle. \quad (\text{B.4})$$

Substituting (B.2) gives the first-order recursion

$$B_{m+1} c_{m+1} = t c_m. \quad (\text{B.5})$$

Since neither t nor any B_m vanishes, specifying one coefficient fixes all the others in both directions. Hence

$$\dim \ker(J_0^- - t) = 1 \quad (\text{B.6})$$

on the completed level-zero space. The same recursion, with an inhomogeneous term, shows that $J_0^- - t$ is surjective on this completion. With the ghost convention of Section 2.5.3, this gives

$$H_{\text{DS,flow}}^0|_{N=0} \cong \mathbb{C}, \quad H_{\text{DS,flow}}^1|_{N=0} = 0. \quad (\text{B.7})$$

Thus the two-sided J_0^3 lattice of the unreduced principal series at fixed s and fixed ϵ produces one level-zero cohomology class. In the main text we set $\epsilon = 0$ throughout. The Virasoro weight itself depends only on the Casimir.

³⁹For the principal series this completion is essential: the Whittaker representative is a formal sum over the zero-mode basis and is not normalizable in the unreduced principal-series norm. But this is not the normalizability condition on the physical state. The physical state is the reduced BRST cohomology class. For real P and $c > 1$, we expect (and find evidence that) the cohomology is the Verma module $\mathcal{V}_{h_P}$. The latter has positive-definite Shapovalov form, with $\langle h_P | h_P \rangle = 1$.

The conformal weight of this class does not have to be inferred from the recursion. The module-independent flowed Hamiltonian was derived in Section 2.5,

$$L_0^{\text{phys}} = L_0^{\text{Sug}} + L_0^{\text{gh}} + \frac{k}{4}, \quad (\text{B.8})$$

so at level zero

$$h = \frac{C_2}{k-2} + \frac{k}{4} = \frac{c-1}{24} + \frac{s^2}{k-2} = \frac{c-1}{24} + P^2, \quad P = \frac{s}{\sqrt{k-2}}. \quad (\text{B.9})$$

The important point is that the J_0^3 dependence has already cancelled in L_0^{phys} . The level-zero representative is spread over the horizontal weight basis, but it nevertheless has a sharp Virasoro weight fixed only by the Casimir.

Using Whittaker/BRST complex methods, we have explicitly checked that the BRST cohomology precisely reproduces the Verma module expectation at levels $N = 0, 1, 2, 3, 4$ for all ghost numbers [33]. The direct calculation gets significantly more complicated at higher levels, compared to the trivial level-0 case we discussed above. We have also checked that $\dim(H^0|_N) = p(N)$, the number of partitions of N , at these levels. In other words, in all sectors examined, the cohomology is non-trivial only at ghost number zero, where its dimensionality matches the Virasoro descendant count. What fails to adapt straightforwardly from the standard theorems on (unflowed) discrete series are the all-level proofs: but they fail irrespective of level, so the fact that we can show that it works by direct cohomology calculations at low levels can be viewed as indicating that it is the proof methods that are failing, and not the reduction itself. We hope to present a complete demonstration in [33]. We also note closely related earlier work by Dhillon [34] and a very recent paper [67]. These papers come from a more mathematical background, and we do not rule out the possibility that the result we want may already be established as a (corollary of a) formal theorem.

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